Designing Stablecoins

Yizhou Cao* Min Dai† Steven Kou‡ Lewei Li§ Chen Yang ¶

Abstract

Existing cryptocurrencies are too volatile to be used as currencies for daily payments. Stablecoins, which are cryptocurrencies pegged to other stable financial assets such as the U.S. dollar, are desirable for payments within blockchain networks, whereby being often called the "Holy Grail of cryptocurrency." By using the option pricing theory and the Ethereum platform that allows running smart contracts, we design several dual-class structures that are written on the ETH cryptocurrency and offer a fixed-income crypto asset (class A coin), a stablecoin (class A' coin) pegged to a traditional currency, and leveraged investment instruments (class B and B' coins). Our investigation of the values of stablecoins in the presence of jump risk and black swan-type events shows the robustness of the design. The design has been implemented on the Ethereum platform.

Keywords: stablecoins, fixed income crypto assets, leveraged return crypto assets, smart contracts.

1 Introduction

How to create a digital currency was a long-standing open problem for many years due to the challenge of duplication and double-spending. A revolution in FinTech is that the two problems can be solved using blockchains. The first blockchain was conceptualized by Nakamoto (2008) and was implemented in 2009 as the core component of the first cryptocurrency, Bitcoin. Another breakthrough came in late 2013 when Vitalik Buterin created the Ethereum platform on which people can write smart contracts that can be tracked and executed automatically. There are currently over 1000 cryptocurrencies traded in exchanges. Some of them are based on public blockchains, such as Bitcoin and Ether (ETH), and others on private blockchains, such as Ripple. One can buy cryptocurrencies from online exchanges or ATMs worldwide, just like buying standard financial securities or foreign currencies.

*Tau Strategy, Singapore 079118, and Research Centre for Quantitative Finance, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong.

†Department of Applied Mathematics, School of Accounting and Finance, and Research Centre for Quantitative Finance, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong.

‡Questrom School of Business, Boston University, 595 Commonwealth Avenue, Boston, MA 02215, USA.

§FinBook, Singapore 048581, and Research Centre for Quantitative Finance, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong.

¶Department of Systems Engineering and Engineering Management, The Chinese University of Hong Kong, Shatin, N.T., Hong Kong.

This paper attempts to design stablecoins, which are cryptocurrencies pegged to other stable financial assets such as U.S. dollars, using concepts from the option pricing theory. Stablecoins are ideal for use as public accounting ledgers for payment transactions within blockchain networks. They also serve as effective crypto money market accounts for asset allocation involving cryptocurrencies.

Background and Motivation. One major drawback of cryptocurrencies is their extreme volatility. Figure 1 shows the price of Ether in U.S. dollars, ETH/USD, from November 9, 2017 to December 31, 2023. During this period, ETH/USD has an annualized volatility of 90.49%, which is more than four times that of the S&P 500 during the same period (20.46%).

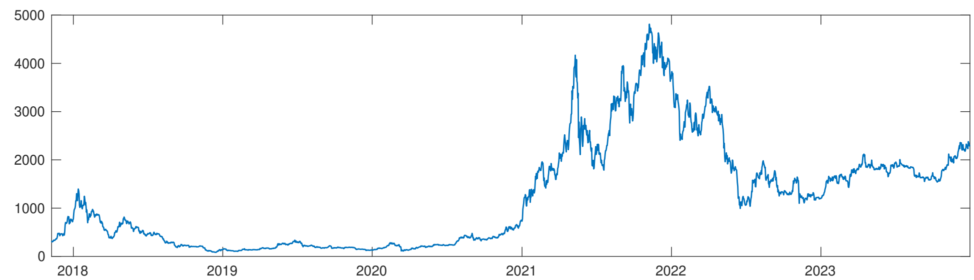

Figure 1: ETH/USD Price from November 9, 2017 to December 31, 2023

The tremendous volatility means that a cryptocurrency like ETH cannot be used as a reliable means to store value. A stablecoin is a crypto coin that keeps stable market value against a specific index or asset, most noticeably the U.S. dollar. Stablecoins are desirable for at least three reasons: (1) They can be used within blockchain systems to settle payments. For example, lawyers or accountants can automatically exchange their stablecoins generated by smart contracts for the services they provide within the system without being bothered by the exchange fees from a cryptocurrency to the U.S. dollar, ranging from 0.7% to 5%. (2) They can be used to form crypto money market accounts and do asset allocation for cryptocurrencies. (3) They can be used by miners or other people who provide essential services to maintain blockchain systems to store values, as converting mined coins into traditional currencies may be expensive.

However, existing cryptocurrencies are too volatile to serve as stablecoins. How to create a good stablecoin is called the "Holy Grail of Cryptocurrency" in the media (Forbes, March 12, 2018; Sydney Morning Herald, February 22, 2018; Yahoo Finance, October 14, 2017). Several examples of the stablecoin design are discussed in Klages-Mundt et al. (2020).

There are three existing types of issuance of stablecoins. The first type is an issuance backed by accounts in real assets such as U.S. dollars, gold, oil, etc. More precisely, these stablecoins represent claims on the underlying assets. For example, the Tether coin is claimed to be backed by USD, with a conversion rate of 1 Tether to 1 USD (see Tether (2016)). However, verifying the claim that Tether has enough reserve in the U.S. dollar is challenging, especially daily.[1] A recent study by

[1] In fact, U.S. regulators issued a subpoena to Tether on December 6, 2017, on whether $2.3 billion of the tokens outstanding are backed by U.S. dollars in reserve (Bloomberg news, 1/31/2018).

Griffin and Shams (2020) shows that Tether is issued without a full backup. In 2018, USDC was introduced to boost transparency by regularly reporting its USD reserves. There are other tokens claimed to link to gold (e.g., Digix, GoldMint, Royal Mint Gold, OzCoinGold, and ONEGRAM), although the claims are also hard to verify.[2]

The second type is the seigniorage shares system, which automatically adjusts the quantity of coin supply and does not necessarily require real assets or cryptocurrency backing. A typical example of this type is the Basis coin (see Al-Naji (2018)). When the coin price is too high, new coins are issued; when the coin price is too low, bonds are issued to remove tokens from circulation. However, maintaining the right balance of supply and demand of a stablecoin is perhaps at the same level of difficulty faced by central banks issuing fiat currencies. Indeed, it was announced in December 2018 that the whole Basis coin project would be shutting down. A more recent attempt in this direction is the algorithmic stablecoin TerraUSD (see Kereiakes et al. (2019)), which aims to relegate the task of maintaining the balance to the market via an arbitrage mechanism. However, TerraUSD de-pegged from \$1 and crashed to \$0.3 on May 11, 2022, due to the limitation of arbitrage in the face of low market liquidity (see Wong et al. (2022) for a comprehensive analysis).

The third type is an issuance backed by over-collateralized cryptocurrencies with automatic exogenous liquidation. For example, one can generate \$100 worth of stablecoins by depositing \$150 worth of Ether. The collateral will be sold automatically by a smart contract if the Ether price reaches \$110. Examples of this type include the DAI token issued by MakerDAO; see MakerTeam (2017). The main drawback of this type is the relatively large collateral size. As pointed out by Klages-Mundt and Minca (2022), even with large collateral, when there is a sudden drop in the collateral value, this type can still be faced with the risk of a "deleveraging spiral" leading to large price volatility.

Our Contribution. The contribution of this paper is threefold: First, we use the option pricing theory and smart contracts to design several dual-class structures that offer entitlements to fixed-income stablecoins (Class A coins) pegged to a traditional currency as well as leveraged investment opportunities (Class B coins). Compared to the existing designs, the stablecoins in our design are backed up by the underlying asset pool consisting of cryptocurrencies rather than real assets. From this perspective, compared with the first type of existing designs above, our design is closer to the other two types, especially the third type. In contrast to the existing designs, whose stability comes from over-collateralization or automatic supply adjustment, our design maintains stability via the split structure and the special upward and downward reset barriers that reduce the credit risk and maintain the stability of future cash flows. More precisely, due to downward resets, a vanilla A coin behaves like a bond with the collateral amount being reset automatically. The A coin can be further split into additional coins, A$'$ and B$'$, where A$'$ behaves like a money market account with a very stable cash flow. The above design can be easily implemented using

[2] In February 2018, the government of Venezuela issued Petro, a cryptocurrency claimed to be backed by one barrel of oil. Recently in April 2020, the Diem Association (formerly known as the Libra Association) announced Libra 2.0 (see Diem Association (2020)), a digital coin backed by a basket of fiat currencies (e.g., USD and the Euro) and low-volatility assets.

smart contracts.[3] A detailed comparison between our stablecoin design and the existing popular designs is provided in Section 2.6.

Second, we show that our proposed stablecoin design is robust and economically viable. (1) The proposed stablecoin A′ can maintain its value stability in the case of black swan-type events (when the underlying cryptocurrency suddenly drops close to zero within an extremely short period), except, for example, when the underlying asset has a sudden drop of more than 60% within a day or an hour (see Section 2.4). (2) The volatility of proposed stablecoin A′ is so low that it is essentially pegged to the U.S. dollar. Table 1 reports the volatility of ETH, S&P 500 index, and gold price denominated in USD, USD index, class A, and A′ coins, under a geometric Brownian motion model. Similar results are also reported in the presence of jump risk (see Section 5).

Table 1: Annualized Volatility of Our Stablecoins versus Common Market Indices

Index	ETH	S&P 500	Gold	USD Index	Class A Coin	Class A′ Coin
Volatility	89.86%	26.09%	15.36%	5.96%	3.48%	1.37%

Annualized volatility is calculated from October 13, 2018 to December 22, 2020.

Third, from a technical viewpoint, in contrast to the standard Black-Scholes partial differential equation (PDE), the pricing equation here is a periodic PDE with a time-dependent upper barrier. We propose a novel iterative numerical procedure (Algorithm 1) and prove its convergence to the solution of the PDE (Theorem 3.2).

Cryptocurrencies have both real usage (e.g., for a transaction) and speculative usage. However, currently, the main cryptocurrencies, such as ETH, serve both purposes, and speculative usage dominates, which makes them too volatile to serve as real usage. A potentially important step is to separate speculation from real usage. This can help the market understand the real usage attribute of cryptocurrencies in addition to speculation, which can potentially help stabilize the market sentiment. Our design aims to separate speculation from the real use of stablecoins via the dual-class structure.

We should emphasize that, compared to the DAI token, our design provides a higher level of robustness in extreme scenarios: In the black swan events when there is a sudden significant drop in the value of underlying cryptocurrency, Class A′ can withstand a larger sudden drop than DAI before taking any loss (see Section 2.4); during a period when the underlying cryptocurrency experiences significant negative returns and the stablecoins face partial liquidation, Class A′ is second-order stochastic dominant over DAI (see Section 7). Furthermore, while users who create DAI are still undesirably exposed to the speculation risk in the underlying cryptocurrency, our design largely separates speculation risk from Class A and A′ (see Section 2.6), which is transferred to Class B instead.

Literature Review. Table 2 summarizes the main differences between our stablecoins and existing popular designs.

[3] The code is on https://github.com/DuoNetwork/duo-contract/blob/0.2.0/contracts/Custodian.sol.

Table 2: Comparison with Existing Stablecoins

	Mechanism	Collateral Asset	Source of Stability	Allowing Tranching	Stable Market Cap
Tether/USDC	Backed by real assets	U.S. dollars	Pegged to U.S. dollar	No	Yes
Libra	Backed by real assets	Basket of low-volatility assets	Pegged to a group of low-volatility assets	No	Yes
Basis	Seigniorage system	N/A	Pegged to U.S. dollar	No	No
DAI Token	Over-collateralization	Ether	Pegged to U.S. dollar	No	No
Our design	Securitization	N/A	Stable future cash flow	Yes	No

Basis (formerly known as Basecoin) project will be shut down, according to an announcement made on Dec. 13, 2018. Stable Market Cap means whether the stablecoins maintain stable market capitalization during the turbulence of the cryptocurrency market. Tether and Libra have a stable market cap since they are directly backed by real assets. The market cap of Basis is not stable, since the seigniorage system reduces (increases) the total supply of the coins when the price of coins drops (increases). DAI token does not have a stable market cap due to the automatic liquidation when the ETH price drops. Our design does not have a stable market cap due to the downward reset when the ETH price drops. Note that although Basis, DAI token, and our stablecoins do not have a stable market cap, they aim at a stable value per coin.

The split structure of our design reminds us of the Collateralized Loan Obligations (CLOs) and the over-collateralization test (O/C test) for CLOs (where Class A and B resemble the senior and equity tranche, respectively). However, to our best knowledge, this paper contributes to the securitization literature by proposing a *much safer way* to do over-collateralization for CLOs via the downward reset mechanism and by incorporating the over-collateralization mechanism directly into the valuation of tranches (resulting in a periodic PDE rather than the standard Black-Scholes PDE).

More precisely, the downward reset differs from the O/C test in the following ways: First, when the O/C ratio is lower than the threshold, the senior tranche payback of the O/C test only requires increasing the ratio back to the threshold, while our downward reset requires increasing it back to the original level (far above the threshold), which potentially reduces the likelihood of repeatedly triggering the downward resets. Second, the reset ratio is monitored in real-time for our downward reset, while it is monitored only monthly for the O/C test of CLOs. Third, we analyze in detail (both with models and with model-free bounds) the benefit of introducing the reset mechanism to the stability of Class A and A' coins. The ratio reset in our design, together with the stability analyses, can potentially be used to safeguard related CLO-like structures.

Many papers discuss the pros and cons of cryptocurrencies. Using cryptocurrencies as a payment method has several benefits. First, as pointed out in Harvey (2016), the core innovation of cryptocurrencies is the ability to publicly verify ownership, instantly transfer ownership, and do that without any trusted intermediary. Thus, cryptocurrencies reduce the cost of transferring ownership. Blockchain technology makes the ledger easy to maintain and reduces the cost of networking (see Catalini and Gans (2016)). The distributed ledger can result in a fast settlement that reduces counterparty risk and improves market quality (see Khapko and Zoican (2018)). Furthermore, since the transaction is anonymously recorded on the blockchain, cryptocurrencies help protect their users' privacy. The underlying technology of cryptocurrencies may strengthen the menu of electronic payment options, while the use of paper currency is phased out (see Rogoff (2015)).

There are also some criticisms of cryptocurrencies. First, a payment system with cryptocurrencies lacks a key feature: the confidence that one can get his money back if he is not satisfied with the goods he purchased. As pointed out by Grinberg (2011), Bitcoin is unlikely to play an essential role in the traditional e-commerce market since consumers typically do not care about the anonymity that Bitcoin provides; instead, they prefer a currency they are familiar with for most goods and services and want fraud protection. Second, Bitcoin does not have identity verification, audit standards, or an investigation system in case something bad happens. For instance, one may lose all his deposit in cryptocurrencies should his password get stolen, and there is no remedy. Furthermore, Blockchain systems are not as trustworthy as they seem to be. Without an intermediary, individuals are responsible for their own security precautions. Finally, it is difficult to value cryptocurrencies.

Despite the significant drawbacks of cryptocurrencies, it is generally agreed that blockchain technologies are here to stay. However, blockchain technologies automatically generate cryptocurrencies to charge the services provided by the system (such as fees incurred by all programming codes that are run on the Ethereum network), crediting essential services to the system (such as the verification services provided by miners), and of exchanging credits for services. Therefore, cryptocurrencies will not disappear as long as blockchain technologies exist. Thus, designing suitable stablecoins is essential for the blockchain system to perform financial functions efficiently and for doing asset allocation across different cryptocurrencies generated by different blockchain systems.

An extensive empirical study of cryptocurrencies is conducted by Chen et al. (2017). Trimborn and Härdle (2018) and Kim et al. (2019) proposed two volatility indices for cryptocurrencies. Hou et al. (2020) study the pricing of options on cryptocurrencies; here, we design nonstandard options with both upper and lower reset barriers, leading to periodic PDEs with a time-dependent upper barrier, which requires a special numerical algorithm and stochastic representations.

The rest of the paper is organized as follows. The design of stablecoins is presented in Section 2. The valuation of these coins is given in Section 3. Numerical examples under the geometric Brownian motion case are given in Section 4. Section 5 extends the geometric Brownian motion framework to a double-exponential jump-diffusion framework. Section 6 discusses the impact of liquidation cost upon downward resets. Section 7 compares our design with DAI in terms of partial liquidation during market downturns. All the technical theorems, propositions, and their proofs are given in the appendix.

2 Product Design

Our design has a dual-class split structure, which, combined with smart contract rules and a market arbitrage mechanism, provides the holders of each class with principal-guaranteed fixed incomes and leveraged capital gains, respectively. More precisely, the Class A Coin behaves like a bond and receives periodic coupon payments, and the Class B Coin entitles leveraged participation of the underlying cryptocurrency, where the initial leverage ratio upon creation is $\frac{1}{1-\alpha}$. Simply put, this split structure means that the holders of Class B coins borrow capital from the holders of Class A coins and invest in a volatile cryptocurrency.

For simplicity, we shall choose ETH as the underlying cryptocurrency[4], and the initial leverage ratio for Class B is set to 2 (i.e., $\alpha = 0.5$); the design of a contract with a general α is discussed in Appendix A. Intuitively, choosing a higher initial leverage ratio (i.e., a larger α) increases the speculative attractiveness of Class B, while choosing a lower initial leverage ratio increases the attractiveness of Class A by making it more stable and less prone to liquidation. A study on the impact of the leverage ratio on the liquidation probability is given in Appendix J.

Our design has great flexibility that enables further splitting. As an example, in this paper we present an extension that further splits Class A into two sub-classes Class A′ and B′ , where one share of Class A′ and B′ jointly invest in two shares of Class A coins. The split structure for Class A′ and B′ resembles that for Class A and B: Class B′ borrows money from Class A′ to invest in Class A. As demonstrated in Sections 4 and 5, Class A′ has a higher level of stability compared to Class A.

2.1 Notations and Formal Definitions

This subsection presents formal definitions of state variables and parameters used in the contract design. A particular design of A and B coins is presented in 2.2. In the following, we use t to denote time and take $t = 0$ as the inception date of all classes.

State Variables. The following variables are tracked by the smart contract at any time t: (1) the relative ETH price P_t/P_0, which is the ratio between the prevailing ETH price in USD P_t, and that on the inception date P_0; (2) the conversion factor β_t used in the creation and redemption process (see below) is defined and updated by the contract, with $\beta_0 = 1$ on inception; (3) the number of days v_t from the inception, last reset, or last regular payout date, whichever is most recent. The relative ETH price is obtained from external price feeds, v is updated daily, and both β and v are updated on payment events through the contract mechanism defined below.

Contract Parameters. Our design includes the following contract parameters: the frequency of regular coupon payout $T > 0$; the coupon rate R and R' for Class A and A′ , respectively (satisfying

[4]From the theoretical perspective, our design is independent of the underlying asset choice and naturally allows the underlying asset to be replaced by a pool of different cryptocurrencies. Doing so can potentially hedge off the idiosyncratic risk within the cryptocurrency pool and lead to a more stable underlying pool value, which further increases the stability of our stablecoin. Furthermore, doing so can help reduce the price risk upon downward resets, provided that cryptocurrencies face liquidity issues. However, the systematic risks, e.g., those driven by common factors (including the risk of black swan events), will likely stay. Therefore, the theoretical model for the underlying ETH can be modified to capture the dynamics of the common factors. When the systematic risk poses a common shock on the values of both the crypto-asset and the U.S. dollar, the impact on the USD value of the crypto-assets is partially offset and is therefore relatively small, since the objective of our design of stablecoins is to achieve stability relative to U.S. dollar. For real implementation, using a pool with multiple cryptocurrencies increases the monitoring cost, e.g., from the price feeding.

$2R > R'$); the subsidy rate $\tilde{R}$ for the design with subsidy from Class A to Class B; the upper bound $\mathcal{H}_u$ and lower bound $\mathcal{H}_d$ (satisfying $\mathcal{H}_u > 1 > \mathcal{H}_d > 0$) used in the upward and downward reset, respectively. The details of the parameters will be specified below.

Creation, Redemption, and Trading. Class A and B coins can be created by depositing ETH shares to a custodian smart contract at any time. Upon receiving two shares of underlying ETH at time t, the custodian contract will return the depositor $\beta_t P_0$ of Class A and Class B coins each. Let $\beta_0 = 1$, which means that two shares of ETH can initially be exchanged for P_0 shares of Class A and Class B each. The conversion factor β_t changes only on payment events, based on the rules to be introduced below. To withdraw ETH at time t, holders of Class A and Class B coins can send, e.g., $\beta_t P_0$ shares of Class A and Class B coins each to the Custodian contract, then the contract will deduct the same amount of Class A and Class B coins, and return the sender two shares of ETH. At any time, two Class A coins can be split into one Class A$'$ coin and one Class B$'$ coin. Conversely, one Class A$'$ coin and one B$'$ coin can be merged into two Class A coins.

Net Asset Values. The net asset values of Class A and Class B shares, to be used in the triggering condition of the reset clauses, can be computed from the state variables automatically without using any models. However, none of the Class A and Class B shares can be directly exchanged for their net asset values. Instead, all classes are traded on exchanges. Thus, the trading prices do not necessarily equal their net asset values but are rather determined by the market in terms of their future cash flows. The market prices will be studied in Section 3 by using the option pricing theory.

At time t, the net asset values of Class A and Class B per share, denoted as V_A^t and V_B^t, are defined as

$$V_A^t = 1 + R \cdot v_t, \tag{2.1}$$

$$V_B^t = \frac{2P_t}{\beta_t P_0} - V_A^t. \tag{2.2}$$

Note that

$$\beta_t (V_A^t + V_B^t) \ = 2P_t / P_0,$$

reflecting the fact that 2 shares of ETH are deposited to form the pool of classes A and B shares, and β_t is the conversion factor determining the number of shares of Classes A and B coins for any new conversion at time t in the pool. The net asset values V_A^t and V_B^t are reference values that represent the stakes of one share of Class A and B in the joint investment pool and determine the timing and payments on the regular payouts and upward/downward resets. Similarly, the net asset values of Class A$'$ and B$'$ are defined as

$$V_{A'}^t \ = 1 + R' \cdot v_t, \quad V_{B'}^t \ = 2(1 + R \cdot v_t) - V_{A'}^t \ . \tag{2.3}$$

Payment Events. Our design includes three types of events triggered automatically by smart contract, which produce cashflows to the classes: (1) the regular coupon payouts upon which Class A, A′ , and B′ receive coupon payments, (2) upward resets upon which all classes receive payments, and (3) downward resets upon which Class A, A′ , and B′ receive payments and are partially liquidated, and all classes undergo a merger. In short, downward resets reduce the default risk of Class B to protect Class A, and upward resets ensure a minimum leverage ratio of Class B to make Class B attractive to leverage investors.

In practice, our design (including the upward and downward reset mechanisms) can be implemented via a custodian smart contract on the Ethereum network via the Solidity language, and they are *automatically* triggered and executed in real-time when the state variables or net asset values satisfy the specified triggering conditions above. See Appendix B for details.

2.2 Vanilla Class A and B Coins

The net asset values of Class A and B coins represent the stakes of Class A and B in the investment pool. More specifically, (2.2) means that each share of Class A and B forms an investment pool with value $\frac{2P_t}{\beta_t P_0}$, within which the stakes of Class A and B are V_A^t and V_B^t, respectively. As a result, during the creation (resp. redemption) of Class A and B coins, users pay (resp. receive) ETH that is worth the pool value.

For example, if the initial ETH/USD price is 500 and $\beta_t = 1$, then two shares of ETH can create 500 shares of Class A coins and Class B coins each, and 500 shares of Class A and B coins each can be redeemed into two shares of ETH. Figure 2 illustrates this split structure.

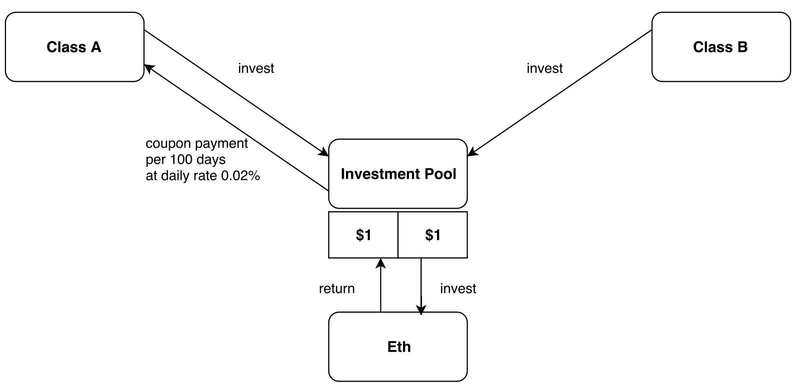

Figure 2: Class A and B, Initial Split. At time t, one share of Class A and B each invests \$1 in ETH. The initial ETH price is $P_0 = \$500$, and the prevailing conversion factor $\beta_t = 1$. So two shares of ETH exchange for 500 shares of Class A coins and 500 shares of Class B coins.

The above design ensures that the initial net asset values of Class A and B coins are both equal to one dollar.

2.2.1 Regular Payout

A regular payout is triggered when the number of days from inception, the last reset, or regular payout reaches T, i.e. $v_{t-} = T$ (or equivalently, $V_A^{t-} = 1 + RT$). On a regular payout, the holder of each Class A coin receives payment with dollar value RT while the holder of Class B coin does not receive any payment.[5] Furthermore, the state variables β and v undergo the changes

$$v_{t+} = 0, \quad \beta_{t+} = \frac{2P_t}{2P_t - \beta_{t-}P_0RT}\beta_{t-}.$$

Subsequently, from the definition of net asset values (2.1) and (2.2), we see that

$$V_A^{t+} = 1, \quad V_B^{t+} = V_B^{t-}.$$

On a regular payout, the accrued interest for Class A is paid out, resulting in its net asset value being reset to 1, and the net asset value of Class B remains unchanged as there is no cash flow for Class B.

For example, assuming $R = 0.02\%$ daily, $T = 100$, and $P_0 = \$500$, a regular coupon payout occurs at time t when $P_t = \$450$ and $\beta_{t-} = 1$, then Class A receives a \$0.02 coupon payment and the conversion ratio is reset to $\beta_{t+} = 1.011$, indicating that two shares of ETH can exchange for 505.62 shares of Class A and 505.62 shares of Class B after the regular payout. This is illustrated in Figure 3.

2.2.2 Upward Reset

An upward reset is triggered when the net asset value of Class B reaches or exceeds the predetermined upper bound, denoted by $\mathcal{H}_u$. On an upward reset at time t, Class A and B's holders will receive payments of value $V_A^t - 1$ and $V_B^t - 1$ per share, respectively. Furthermore, β and v are updated as

$$v_{t+} = 0, \quad \beta_{t+} = \frac{P_t}{P_0}.$$

Subsequently, the net asset values become

$$V_A^{t+} = V_B^{t+} = 1.$$

Upon an upward reset, both the accrued coupon for Class A coins and the leveraged profit from ETH appreciation for Class B coins are paid out, and, therefore, their net asset values are reset to 1. Also, the conversion factor β_t resets so that two shares of ETH can be exchanged for P_t shares of Class A and B after the upward reset.

For instance, as illustrated in Figure 4, after another 50 days, assuming the ETH price gradually

[5]Payments are made in the form of underlying ETH from the Custodian contract. For instance, upon regular payout, the holder of each Class A coin receives underlying ETH with the amount $\frac{RT}{P_t}$.

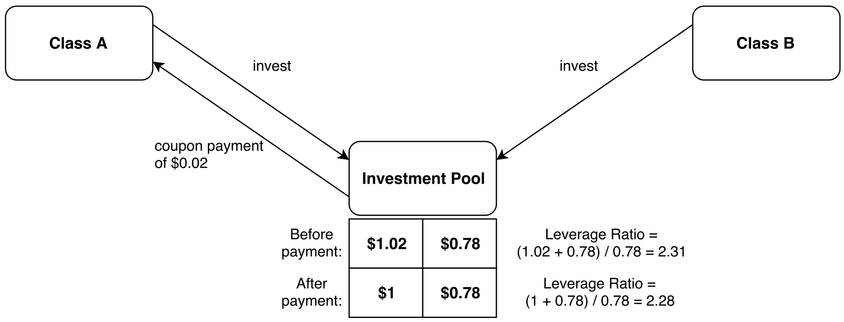

Figure 3: Class A and B, Regular Payout. After 100 days, the ETH price drops to \$450, so the total investment of one Class A coin and one Class B coin becomes \$1.8, within which \$1.02 belongs to Class A. A regular payout takes place, and Class A receives a \$0.02 coupon payment. New exchange ratio: 2 shares of ETH now correspond to 505.62 ($= 500 \times \frac{2\times 450}{2\times 450 - 500 * 0.02}$) shares of Class A and 505.62 shares of Class B, yielding $\beta_{t+} = 1.011$.

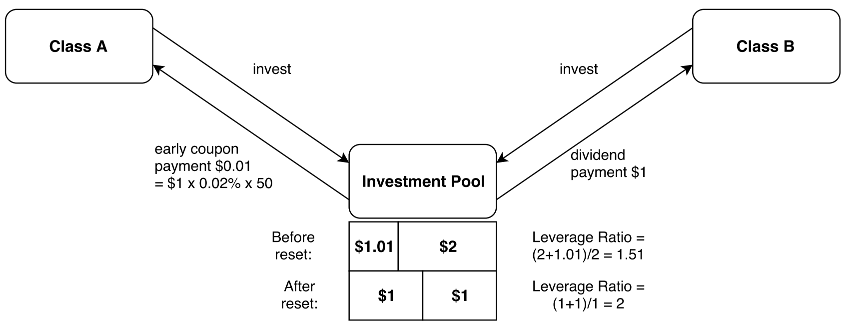

Figure 4: Class A and B, Upward Reset. After 50 days, the ETH price grows to \$760.96, and Class B NAV grows to \$2, triggering an upward reset. Class A NAV equals \$1.01, where \$0.01 is the 50-day accrued coupon. On this date, Class A receives a \$0.01 coupon payment, and Class B receives a \$1 dividend payment. New exchange ratio: 2 shares of ETH now correspond to 760.96 shares of Class A and 760.96 shares of Class B, yielding $\beta_{t+} = 760.96/500 = 1.52$.

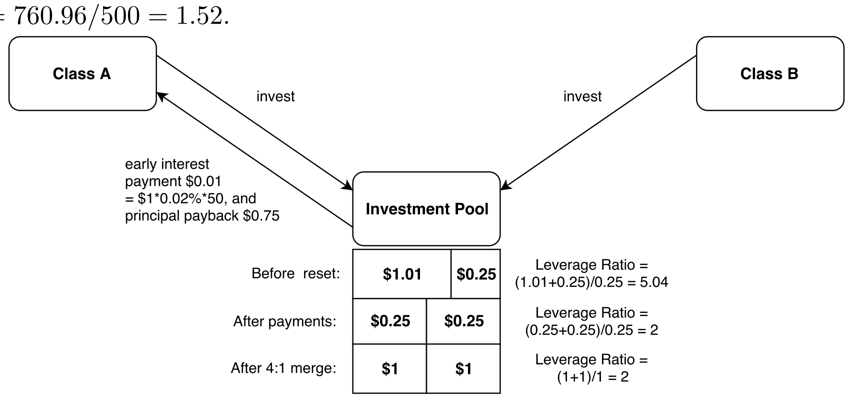

Figure 5: Class A and B, Downward Reset. After another 50 days, the ETH price drops to \$479.40, and Class B NAV drops to \$0.25, triggering a downward reset. Again, Class A NAV equals \$1.01, where \$0.01 is the 50-day accrued coupon. On this date, Class A receives a \$0.01 coupon payment, as well as a \$0.75 principal payback. Then, Class A and B each undergo a 4:1 merger, so both have NAV equal to \$1. New exchange ratio: 2 shares of ETH now correspond to 479.40 shares of Class A and 479.40 shares of Class B, yielding $\beta_{t+} = 0.96$.

grows to \$760.96, so the net asset value of Class B grows to $\mathcal{H}_u \equiv \$2$, triggering an upward reset. The holders of Class A and B receive payments in the amounts of \$0.01 and \$1, respectively. Two shares of ETH now correspond to 760.96 shares of Class A and 760.96 shares of Class B, yielding $\beta_{t+} = 1.52$.

2.2.3 Downward Reset

A downward reset is triggered when the net asset value of Class B coins reaches or drops below the predetermined lower bound, denoted by $\mathcal{H}_d$, i.e. $V_B^t \leq \mathcal{H}_d$. On a downward reset at time t, Class A holders receive payment with dollar value $V_A^t - |V_B^t|$ per share, and Class B holders receive no payment. The state variables β and v are updated as

$$v_{t+} = 0, \quad \beta_{t+} = \frac{P_t}{P_0}.$$

Furthermore, each share of Class A and B is reduced to $(V_B^t)^+$ share, and as a result,

$$V_A^{t+} = V_B^{t+} = 1.$$

There are two scenarios for the downward rest. First, if $0 < V_B^t \leq \mathcal{H}_d$, a downward reset is triggered when the pool value (per share of Class A and B) $\frac{2P_t}{\beta_t P_0} = V_A^t + V_B^t > V_A^t$. In this case, (i) Class A receive the accrued coupon payment Rv_t, which reduces its net asset value to 1; (ii) a fraction $1 - V_B^t$ of Class A is liquidated at its net asset value, resulting in another payment $1 - V_B^t$ per share and a remaining net asset value of V_B^t (the same as the net asset value of Class B); (iii) Each $1/V_B^t$ shares of Class A and B are merged into one share, respectively, which increase the net asset values of both Class A and B to 1. Note that the two parts of payment to Class A sum up to $V_A^t - V_B^t$ due to (2.1). Also, in this scenario, the leverage ratio reverts to the initial value of 2. Furthermore, the conversion factor β_{t+} resets to P_t/P_0, meaning that two ETHs can exchange for P_t shares of Class A and B after the reset.

Second, if $V_B^t \leq 0$, a downward reset is triggered when the pool value $\frac{2P_t}{\beta_t P_0} \leq V_A^t$; in other words, the pool value is no longer greater than the net asset value of Class A. The above mechanism of downward reset means that Class A holders receive all the remaining pool value $\frac{2P_t}{\beta_t P_0} = V_A^t + V_B^t = V_A^t - |V_B^t|$, and each share of Class A and B is reduced to $(V_B^t)^+ = 0$ share, meaning that Class A is fully liquidated. This implies that when $V_B^t < 0$, Class A holders receive a liquidation value lower than the net asset value and, therefore, take a loss. Note that after the full liquidation, users can again create new Class A and B via the creation rule specified in Section 2.1.

By (2.2), the downward reset is triggered by reductions in ETH price P_t. If a jump in the ETH does not occur, then any downward reset would be triggered exactly at $V_B^t = \mathcal{H}_d$; in other words, the net asset value of Class B reaches the lower bound rather than jumping below it. In this case, according to the downward reset rule described above, Class A holders receive payment with dollar value $V_A^t - \mathcal{H}_d$, and $1/\mathcal{H}_d$ shares of Class A and B are merged into one share of Class A and B, respectively. For instance, as illustrated in Figure 5, assuming after another 50 days, the ETH price gradually drops to \$479.40, so the net asset value of Class B drops to $\mathcal{H}_d \equiv \$0.25$, triggering a downward reset. Class A receives a \$0.01 coupon payment and a \$0.75 principal payback, and then both classes undergo a 4:1 merger. Two shares of ETH now correspond to 479.40 shares of Class A and 479.40 shares of Class B, yielding $\beta_{t+} = 479.40/500 = 0.96$.

Under a black swan event of the ETH, the net asset value of Class B coins V_B^t is likely lower than $\mathcal{H}_d$ or even becomes negative upon a downward reset, due to the sharp drop in ETH price. Note that it is assumed that when a time-t reset is triggered by a jump in V_B, it takes place immediately *after* the jump. To differentiate, in the description above, we assume that P and V_B jump between $t-$ to t, while the reset occurs between t to $t+$. For instance, consider a time-t downward reset triggered by a downward jump in V_B such that $V_B^t < \mathcal{H}_d \leq V_B^{t-}$. V_A and β maintain their value across the jump, i.e., $V_A^{t-} = V_A^t$ and $\beta_{t-} = \beta_t$, which can be different from the value V_A^{t+} and β_{t+} respectively, due to the reset. Also, the payment of Class A depends on V_A^t and V_B^t. This subtle difference between the changes in P and V_B and the changes in V_A and β is because the former is caused by an external shock, while the internal reset mechanism causes the latter. This assumption is reasonable since, in practice, the reset can only take place after observing the jumps, no matter how fast the system reacts to the jump.

2.2.4 Discussions on the Design of Class A and B

Since Class A and B shares are traded on exchanges, besides their net asset values, they also have market/model prices, which will be studied in Section 3. *No arbitrage* implies that the market prices of Class A and B coins also satisfy the parity relation:

$$W_A^t + W_B^t = \frac{2P_t}{\beta_t P_0}, \tag{2.4}$$

where W_A and W_B are the market prices of the Class A and B coins, respectively. This has an interesting implication: Suppose the demand for Class B coins is low, then the market value of Class B coins would be low, and the above parity relation implies that the market value of the Class A coins would be high.

The above split design is related to the tranching technique in securitization (e.g., collateralized debt obligations). What is new here is the carefully designed reset mechanism. Without this reset mechanism, tranching alone is not enough to control the risk of Class A coins (and A′ to be introduced in Section 2.3), especially when the underlying value becomes low. For example, the downward reset mechanism reduces the risk by partially paying back the principal when the underlying price is low. As a result, upon a reset, the tranches are restructured based on their value before the reset and a pre-specified set of rules.

The incorporation of the reset mechanism requires a delicate treatment even for a single tranche; one cannot apply existing financial engineering results to solve for the value of each tranche. Indeed, the future cash flow of Class A coins is periodic and path-dependent on the underlying value, resulting in a periodic PDE rather than the standard Black-Scholes PDE. To our best knowledge, this type of cash flow has not been extensively discussed in securitization literature. In Section 3, we derive a rigorous PDE characterization of the fair value of this type of cash flow under a geometric Brownian motion assumption, enabling a numerical solution via PDE.

Class A coin behaves like a corporate bond. Its coupon payment is sustained by the underlying investment pool. Therefore, from the perspective of Class B, the coupon payment can be viewed as the cost of borrowing from Class A. As a result, the coupon rate R can be used to adjust the relative attractiveness of Class A and Class B coins. Although Class A has a fixed coupon rate and its coupon payment is periodic and protected by the downward resets, its value is still volatile on non-coupon dates. This is because the coupon rate is usually higher than the risk-free rate, and on a downward reset, a portion of Class A coin will be liquidated, and then the holder of Class A will lose high coupons that would be generated from this portion. Therefore, a potential downward reset will make the price of Class A volatile. In Section 2.3, we will propose one more type of stablecoins: A′ coins.

The above design of A and B coins is partially motivated by the dual-purpose fund in China[6], and the main differences are summarized as follows. First, upon upward and downward resets and regular payouts, to maintain the no-arbitrage relation, our design adjusts the exchange ratio of the shares between the underlying ETH and Class A and B coins, unlike the dual-purpose funds which adjust the underlying fund, since the stablecoins have no direct control over the underlying ETH. Second, the triggering condition of the upward reset of our stablecoin is based on the net asset value of Class B rather than the underlying fund value. Third, the underlying funds of Chinese dual-purpose funds incur management fees, whereas the underlying ETH does not. Finally, the periodic payout of dual-purpose funds is annually at a fixed date, while the periodic payout of our Class A coins happens when a pre-specified time has passed from the last reset or payout event. This reduces the frequency of payouts, making the coins more stable.

One might worry about the demand for Class B coins, as they are leveraged products. If the demand for Class B coins is low, then the price of Class A coins may be overpriced. To provide extra incentive for investors to hold Class B coins, one can modify the contract so that Class B coin holders receive coupon payments from Class A coins. The details are given in Section 2.5.

2.3 Class A′ and B′ Coins

In this design, Class A′ and Class B′ jointly invest in two shares of Class A. The split structure for Class A′ and B′ resembles that for Class A and B in that Class B′ borrows money from Class A′ at the rate R' to invest in Class A. Here R' is set to be close to the risk-free rate r, whereas the rate R for Class A is generally much higher.

The resets or regular payouts of Class A′ and B′ are triggered *when and only when* Class A resets or gets a regular payout, respectively. On a regular payout, Class A′ and B′ receive payments of

[6]See Dai et al. (2018); for related dual-purpose funds in the U.S., see Ingersoll (1976) and Jarrow and O'Hara (1989).

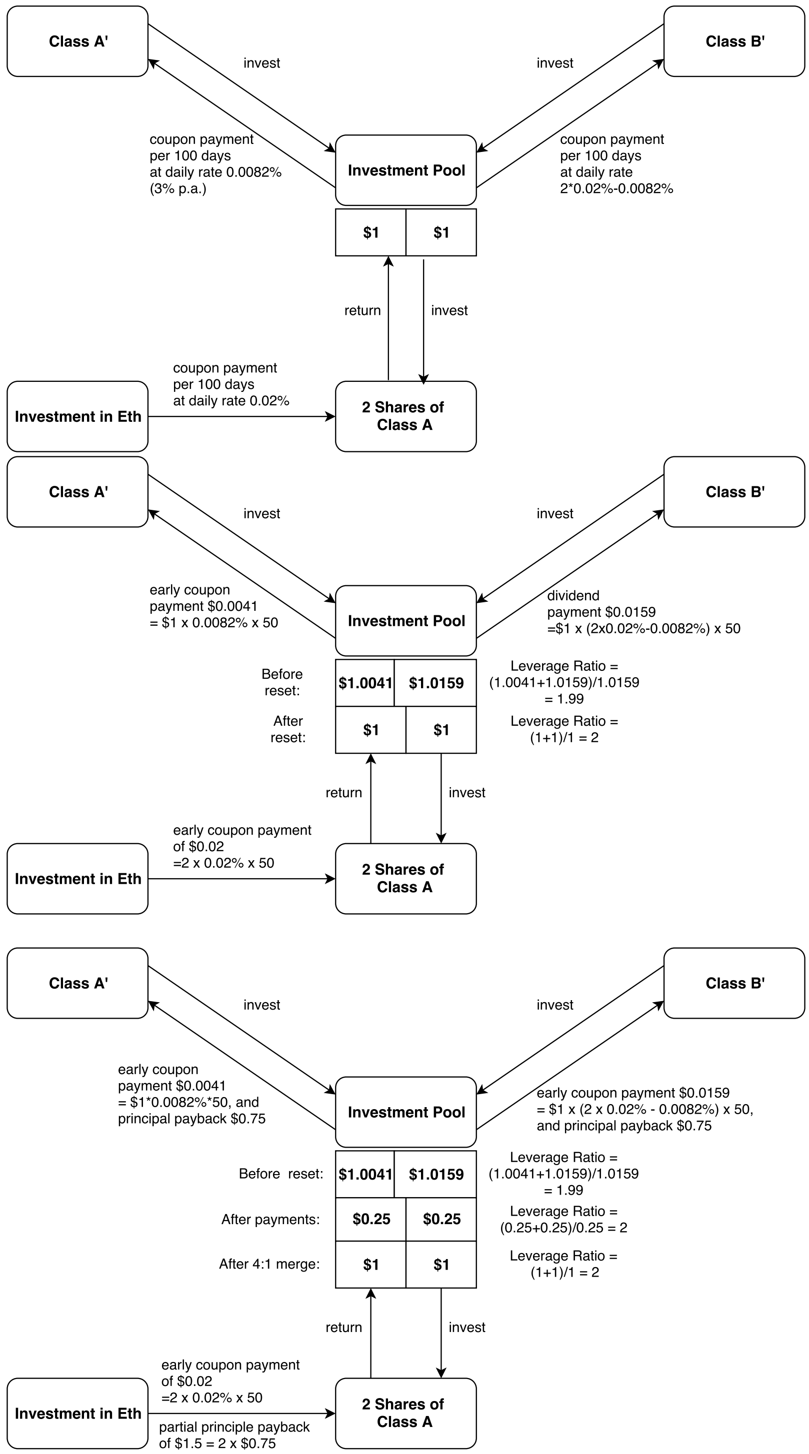

Figure 6: Top Figure: What happens to Class A′ on a regular payout date of A. On regular payout dates for Class A (per 100 days), two shares of A receive a coupon payment of \$0.04, i.e., at a daily rate of 0.02%. \$0.0082 is paid to A′ and \$0.0318 to B′ . **Middle Figure:** Upward Reset of Class A′ . After 50 days, Class B's net asset value grows to \$2, triggering an upward reset. **Bottom Figure:** Downward Reset of Class B′ . After another 50 days, Class B's net asset value drops to \$0.25, triggering a downward reset.

$R'T$ and $(2R - R')T$, respectively. On an upward reset at time t, Class A$'$ and B$'$ receive payments of $V^t_{A'} - 1$ and $V^t_{B'} - 1$, respectively (see (2.3) for the definition of $V^t_{A'}$ and $V^t_{B'}$). On a downward reset at time t, each share of both Class A$'$ and B$'$ is reduced to $(V^t_B)^+$ share. Furthermore, Class A$'$ receive a payment of $\min\{V^t_{A'} - (V^t_B)^+, 2(V^t_A + V^t_B)\}$, and Class B$'$ receive a payment of $\min\{V^t_{B'} - (V^t_B)^+, 2(V^t_A - |V^t_B| - \min\{V^t_{A'} - (V^t_B)^+, 2(V^t_A + V^t_B)\})\}$.

The payments on a downward reset can be understood in the following scenarios: (i) If $V^t_B > 0$, then Class A$'$ and B$'$ receive $V^t_{A'} - V^t_B$ and $V^t_{B'} - V^t_B$, whose summation, according to (2.3), is $2(V^t_A - V^t_B)$ which is exactly the payment to two shares of Class A (see Section 2.2.3); (ii) If $V^t_B \leq 0$ and $2(V^t_A + V^t_B) \geq V^t_{A'}$, then according to Section 2.2.3, two shares of Class A receives $2(V^t_A + V^t_B)$ which is split between Class A$'$ and B$'$; Class A$'$ receives its net asset value $V^t_{A'}$ and Class B$'$ receives the remaining value of $2(V^t_A + V^t_B) - V^t_{A'}$; (iii) If $V^t_B \leq 0$ and $2(V^t_A + V^t_B) < V^t_{A'}$, then the payment to two shares of Class A is insufficient to cover the net asset value of Class A$'$, and therefore Class A$'$ receive the entire payment $2(V^t_A + V^t_B)$ while Class B$'$ receives nothing.

Using the same example as in Section 2.2, Figure 6 illustrates the cash flow of Class A$'$ and B$'$ coins under gradual ETH price changes.

As we will see later, Class A$'$ behaves like a money market account, except in the extreme case in which the underlying asset suddenly suffers huge losses in a very short period (e.g., a sudden 60% or more drop within a day or an hour). Here is the intuition on this. The stability of Class A$'$ coin price comes from its stable future cash flow, as Class A, B, and B$'$ coins act like buffers for Class A$'$. Indeed, the volatility of ETH spreads to Class A$'$ coin only via the timing of coupons (on regular payment or upward resets) or the partial payback on principal (on downward reset dates). Since the accrued amount of the coupon grows linearly with respect to time, and the coupon is expected to be paid at least once per 100 days, the timing of the coupon payment does not cause large volatility on Class A$'$ nominal price; the volatility of the total return (taking into account the coupon) is even smaller. By setting the coupon rate to be close to the risk-free rate, the value of future coupons is close to 1. Therefore, on a downward reset date, the partial payback of the principal (at book value) also does not cause a large impact on the cash flow.

2.4 A Model-Free Bound for Black Swan Events

The cryptocurrencies are subject to *black swan events*, in which their values may jump down sharply. Despite the protection provided by the downward reset mechanism, when there is a sudden drop in ETH value Class A and A$'$ can still incur a loss, i.e., losing part of the accrued coupon or even the principal. Here we give a model-free bound on how large the jump in ETH should be to trigger such a loss.

We take Class A$'$ as an example. According to the contract design, on a downward reset time t, Class A$'$ takes a loss if and only if the net asset value of Class B becomes negative, i.e., $V^t_B \leq 0$, and the payment to Class A$'$ (i.e., the net asset value of 2 shares of Class A) is lower than the net asset value of Class A$'$, i.e., $2(1 + Rv_t - |V^t_B|) < 1 + R'v_t$. These two conditions are equivalent to $V^t_B \leq 0$ and $V^t_B < \frac{R'v_t - 1}{2} - Rv_t$. In other words, assuming $(R' - 2R)T - 1 \leq 0$, Class A$'$ takes a loss if and only if V_B *jumps* from $\mathcal{H}_d$ or above to a negative level below $\frac{R'v_t-1}{2} - Rv_t$ (that is, $V_B^{t-} \geq \mathcal{H}_d$ and $V_B^t \leq \frac{R'v_t-1}{2} - Rv_t$; if V_B drops downward without a jump, a downward reset will be triggered instead, and Class A$'$ will not take a loss). This further means that, assuming $(R'-2R)T - 1 \leq 0$, to incur a loss in Class A$'$ coins, the ETH price must have a (negative) return lower than $\frac{1}{2}\frac{R'v_t+1}{Rv_t+1+\mathcal{H}_d} - 1$.

In our design example, we set $R = 7.3\%$ p.a., $R' = r = 3\%$ p.a., $\mathcal{H}_u = 2$, $\mathcal{H}_d = 0.25$, $T = 100$ days. With these contract parameters, we have

$$\max_{0 \leq v \leq T} \frac{1}{2}\frac{R'v+1}{Rv+1+\mathcal{H}_d} - 1 = -60\%,$$

meaning that unless there is a sudden downward jump of more than -60% between two adjacent monitoring time points, Class A$'$ will not take a loss. In comparison, the historical maximal single-day loss of ETH is recorded at -72.82% on its second trading day (Aug 8, 2015), at which time the market was yet to be familiar with it. The maximum single-day loss thereafter is only -43.88% recorded on March 12, 2020, which is not large enough to trigger a loss in the Class A$'$ coin. In our implementation, the ETH price is monitored at an hourly frequency. In our hourly dataset from October 13, 2017 to December 22, 2020, the largest recorded single-hour loss is -19.82% at 10:00 of March 12, 2020. Therefore, Class A$'$ would take a loss only when the ETH price jumps downwards by 60% or more within an hour, which is even less likely.

In comparison, the DAI token starts to take a loss when its total collateral value suddenly drops from 150% of the DAI value to below 100%, i.e., with a -33% downward jump. Therefore, the Class A$'$ coin may withstand a larger sudden downward jump in ETH than the DAI token. By further splitting Class A$'$ to get A$''$, the resulting new stablecoin A$''$ can withstand even larger downward jumps before taking any loss.

2.5 A Contract Design with the Subsidy from Class A Coins to Class B Coins

One might worry about the demand for Class B coins, as they are leveraged products. If one wants to provide an extra incentive for investors to hold Class B coins, one can modify the contract so that Class B coin holders receive coupon payment from Class A coins.

The modified design with subsidy from Class A to B differs from the one introduced in Section 2.2 and 2.3 only in the payoff on downward resets. Specifically, on a downward reset time t, if $V_B^t > 0$, Class A and B holders receive payment with dollar value[7] $V_A^t - V_B^t - (V_A^t - 1)\frac{\tilde{R}}{R}$ and $(V_A^t - 1)\frac{\tilde{R}}{R}$, respectively, and each share of Class A and B is reduced to V_B^t share of Class A and B, respectively. Class A$'$ holders still receive an accrued coupon payment $R'v_t$ plus the value $1 - V_B^t$ of the liquidated shares. Class B$'$ holders receive a coupon payment $(2R - R')v_t$ plus the value $1 - V_B^t$, *minus* $2(V_A^t - 1)\frac{\tilde{R}}{R}$; and then each share of both Class A$'$ and B$'$ is reduced to V_B share.

[7]This equals $V_A^t - (V_B^t)^+ - \tilde{R}v_t$; recall that v_t is the time from the inception, last reset or regular coupon payout, as defined in (A.3). We require $0 \leq \tilde{R} \leq \frac{1-\mathcal{H}_d}{2T} + R - \frac{R'}{2}$ to ensure that Class A, B, A$'$ and B$'$ all receive nonnegative payments in normal stances where $V_B \geq 0$. Under the contract parameter given in Section 2.4, this means $0 \leq \tilde{R} \leq 0.387\%$ (142.67% p.a.).

In extreme events the net asset value of Class B can jump to a nonpositive value, that is $V_B^t \le 0$. In this case, all four shares are fully liquidated. Class A receives a payment with dollar value $(V_A^t - |V_B^t| - (V_A^t - 1)\frac{\tilde{R}}{R})^+$, and Class B receives $\min\{(V_A^t - 1)\frac{\tilde{R}}{R}, V_A^t - |V_B^t|\}$. That is, Class B receives the full subsidy if Class A can still cover it; otherwise, Class B receives the remaining net asset value of Class A, if any. Class A′ receives its full asset value $1 + R'v_t$, or the remaining total asset for A′ and B′ , $2(V_A^t - |V_B^t| - (V_A^t - 1)\frac{\tilde{R}}{R})^+$, whichever is smaller. Finally, Class B′ receives the remaining total asset value for A′ and B′ minus the payment to A′ .

In this case we can also get a model-free bound on how large the jump in ETH should be to trigger such a loss. Class A′ will take a loss when and only when $V_B^t \le 0$ and $2(V_A^t - |V_B^t| - (V_A^t - 1)\frac{\tilde{R}}{R})^+ < 1 + R'v_t$, i.e., the payment to Class A′ is lower than the net asset value of Class A′ , on a downward reset date. A similar argument as in Section 2.4 shows that assuming $(R' - 2R + 2\tilde{R})T - 1 \le 0$, this will happen only when the ETH price has a sudden (negative) return lower than $\frac{1}{2}\frac{R'v_t+1+2\tilde{R}v_t}{Rv_t+1+\mathcal{H}_d} - 1$. Using the contract parameter as in Section 2.4 as well as $\tilde{R} = 10\%$ p.a., we have

$$\max_{0 \le v \le T} \frac{1}{2}\frac{R'v_t + 1 + 2\tilde{R}v_t}{Rv_t + 1 + \mathcal{H}_d} - 1 = -52.4\%,$$

i.e., unless there is a sudden downward jump of more than -52.4%, Class A′ will not take a loss.

2.6 Differences between Our Design and the Existing Popular Designs

We now compare our stablecoin design with the existing popular designs, Tether, USDC, and DAI, from the perspectives of issuers, users, and regulators, as summarized in Table 3.

Comparison from Issuers' Perspective

An issue arises for real asset-backed stablecoins such as Tether and USDC: How can a user verify the deposit account in real time without asking an accountant to check it? From the issuer's perspective, this issue makes it harder to convince users of the trustworthiness of the real asset reserve behind the stablecoins. In contrast, our design and DAI have a greater level of transparency because the market value of the coin deposit can be observed instantaneously as one wishes.

The main difference between our design and the DAI token is that DAI needs a relatively large collateral size since it adopts issuance backed by over-collateralized cryptocurrencies with automatic exogenous liquidation, while our design does not need this since it creates stablecoins by using tranches with carefully designed upward and downward resets. More precisely, to generate \$100 worth of stablecoins, the DAI design needs to deposit at least \$150 worth of ETH, meaning that \$50 is locked away from circulation. For our design, the firm mainly serves to do the conversion, with all the A, B, A′ , and B′ coins traded outside the firm, and no value is locked away. Thus, the capital requirement of our design may be much less than that of the DAI token.

Another difference lies in the fluctuation of the profits and losses for issuing firms and maintenance costs. The issuing firms of stablecoins backed by real assets may face the risk of price fluctuation of real assets (e.g., gold and oil). For Tether and USDC, there is a significant capital cost associated with maintaining the capital reserve. For the DAI token, the main profit comes from the stability fee, that is, the loan fee paid by the DAI users for over-collateralization through the DAI deposit, although there is a large capital requirement. Similarly, the issuer of our design can charge a conversion service fee between the underlying token and Class A and B, without facing the fluctuation of the real asset prices. More importantly, the main job of the issuer of our design is to convert between different classes of assets and ETH via smart contracts (not to maintain capital reserve).

Comparison from Users' Perspective

Next, we switch to the users' perspective. Compared with the DAI token, our design largely separates speculation from the stablecoins (Class A and A$'$). To wit, consider a period $[T_1, T_2]$, for which a user creates a stablecoin (either Class A or DAI token) at T_1 and redeems it at T_2. Further, assume that the underlying ETH price increases during this period. Users who create DAI still benefit from the appreciation in ETH (as collateral) value during this period, since when DAI is returned and ETH is unlocked from the custodian contract at T_2, its price is higher than at T_1. In contrast, users who create Class A barely benefit from the appreciation in ETH: At T_1, they pay the market value of Class A coins (by converting ETH into both Class A and B coins and then immediately selling the Class B coins at their market price), and at T_2 they receive again the market value of Class A coins (by buying Class B coins at their market price and converting Class A and B into ETH). Since the market value of Class A coins is close to 1 at both T_1 and T_2, the users barely obtain any profit despite the ETH price increase. Instead, users who create and redeem Class B coins benefit from such an increase. In return for giving up their benefit from ETH appreciation, the stablecoin users in our design, compared to DAI users, do not risk losing value in the extra collateral and receive interest payments instead of having to pay loan interest in the form of a stability fee.

When the value of the collateral (for DAI and our design) or reserve asset (for Tether and USDC) drops, the stablecoin users are faced with potential losses. DAI users benefit from the appreciation in ETH as just described, but they will also take the first loss from a drop in ETH price, as they get a lower value from the unlocked ETH when they redeem DAI. In particular, when the value is sufficiently lower, users pose more ETH deposits or redeem DAI to reduce leverage. As studied by Klages-Mundt and Minca (2022), the latter may lead to a "deleveraging spiral" that increases the volatility of DAI. In contrast, for Tether and USDC, it is the issuers who take the first loss in terms of their reserves, and users only start to take a loss when the drop is so large that the coins are facing debasement.

Our design is closer to Tether and USDC in this sense. When the ETH price drops, the holders of the speculative Class B take the first loss. When the loss is large enough to trigger a downward reset, Class A shares part of the loss by forgoing part of future interest payments in a partial liquidation. Only in the extreme black swan events does Class A start to lose part of its principal.

From the stablecoin users' perspective, the issuance efficiency of stablecoins in our design is higher than that of DAI tokens. Although generating Class A and A′ coins via splitting will generate Class B and B′ coins, the latter can be sold off on the exchange. In this way, the stablecoin users can create Class A and A′ coins at their market value without requiring additional value as collateral.

There are also two technical differences between our design and the DAI token. First, they are different financial instruments. By design, DAI is an instrument pegged to the U.S. dollar and essentially a loan secured with over-collateralization through an auto-liquidation mechanism when the collateral-to-debt ratio drops below a certain threshold (150% at present). In contrast, Class A and A′ coins are more like a bond and a money market account, respectively. The stability of their values comes from the stability of their future cash flows. Second, compared with DAI, our design has an additional feature of tranching with carefully chosen reset barriers to reduce risk and enhance stability. As illustrated in Section 2.3, Class A coins can be split into A′ and B′ coins. Since Class A′ coins are backed by both Class B and B′ coins, they are more stable and more robust to black swan events than Class A coins, as will be demonstrated numerically in Section 4. Based on the same idea, one can further split Class A′ to get an even stabler type of coins, etc. It should be noted that the tranche structure in our design is not standard; when the value of the Class B tranche falls below the lower threshold to trigger the downward reset, the remaining tranches are restructured and partially paid back.

Being the stablecoin with the largest market cap (according to coinmarketcap.com), Tether has been relatively stable over a long period, even after the revelation in April 2019 that it was not fully backed up (see https://www.coindesk.com/markets/2019/04/30/tether-lawyer-admits-stablecoin-now-74-backed-by-cash-and-equivalents/). Still, as pointed out by Li and Mayer (2021), the risk of debasement remains. The DAI token, on the other hand, experienced "Black Thursday" on March 12, 2020, due to the cascade of liquidation caused by the ETH price drop, although it has been relatively stable since late 2020. As for our design, its stability will be illustrated numerically in Section 4, demonstrating that the resulting volatility is indeed small, especially for Class A′ coins almost pegged to USD.

Comparison from Regulators' Perspective

Thanks to the use of smart contracts, the collateral asset in our design and the DAI token can be verified by regulators in almost real-time, and therefore they offer a much higher level of transparency than Tether and USDC (even if their real reserves are monthly reported).

3 Valuation

3.1 Class A and B coins

Denote the relative underlying value to be $S_t = P_t/(\beta_t P_0)$. The parity relation (2.4) between the market value of Class A coins W_A^t and B coins W_B^t can be rewritten as $W_A^t + W_B^t = 2S_t$, which allows us to focus on the valuation of Class A coins.

Table 3: Comparison with Existing Stablecoins

	Issuing Firms	Users	Regulators
Tether USDC	⊕ Larger profits (and loses) from maintaining reserve	⊕ Empirically stable	
	⊖ Need to convince users of the trustworthiness of reserve	⊖ Risk of debasement	⊖ Lack of transparency; coins having the risk of not being fully backed up by real assets
DAI	⊕ Ease of management due to use of smart contract	⊕ Empirically stable (although less so than Tether/USDC)	⊕ Transparency due to use of smart contract
	⊖ Inefficient issuance due to excess risky collateral	⊖ Users take first loss (in collateral margin) ⊖ Risk of full liquidation and even loss ⊖ Stability fee	
Our Design	⊕ Ease of management due to use of smart contract ⊕ Efficient issuance as no excess collateral required	⊕ Various tranche suit the risk appetite of different investors ⊕ Separating speculation from real usage ⊕ Stablecoin users receive interest	⊕ Transparency due to use of smart contract
	⊖ (smaller) profit for the issuers mainly from conversion fees	⊖ Risk of partial liquidation and even loss (but with lower probability than DAI)	

This table compares our design of stablecoins with the existing designs of Tether, USDC, and DAI, by listing their pros (marked by ⊕) and cons (marked by ⊖) from the perspectives of issuing firms, users, and regulators, respectively.

Since the future cash flow of the coins depends on the cumulative time from the last payment (namely, last reset or last regular payout), rather than the calendar time, in the remainder of this section we shall relabel the last payment time as 0, so that $t \in [0, T]$ denotes the time from the last payment. Denote by $H_d(t) = \frac{1}{2}(1 + Rt) + \frac{1}{2}\mathcal{H}_d$ and $H_u(t) = \frac{1}{2}(1 + Rt) + \frac{1}{2}\mathcal{H}_u$ the downward reset barrier and the upward reset barrier (associated with S_t), respectively. Hence, the downward (respectively, upward) reset is triggered when S_t hits $H_d(t)$ (respectively, $H_u(t)$). Note that the upward reset barrier H_u is time-dependent, in contrast to the dual-purpose fund in China, which has a constant upward reset barrier. It is easy to see $H_d(t) \leq S_t \leq H_u(t)$ if S_t changes continuously. By design, S returns to 1 on every reset date and is reduced by $\frac{1}{2}RT$ on every regular payout date.

The market value of Class A coins, $W_A^t \equiv W_A(t, S)$, is given recursively as

$$W_A(t,S) = E_t\Bigg[e^{-r(T-t)}[RT + W_A(0, S_T - RT/2)] \cdot \mathbf{1}_{\{T<\tau\wedge\eta\}} + e^{-r(\tau-t)}[R\tau + W_A(0,1)] \cdot \mathbf{1}_{\{\tau\leq T\wedge\eta\}}$$
$$+ e^{-r(\eta-t)}[R\eta + 1 - |V_B^\eta| + (V_B^\eta)^+ W_A(0,1)] \cdot \mathbf{1}_{\{\eta\leq T\wedge\tau\}}\Bigg], \quad (3.1)$$

where r is the risk-free rate, random times τ and η represent the first upward and downward reset date from t, respectively (i.e., $\tau = \inf\{\xi \geq t : S_\xi \geq H_u(\xi)\} \equiv \inf\{\xi \geq t : V_B^\xi \geq \mathcal{H}_u\}$ and $\eta = \inf\{\xi \geq t : S_\xi \leq H_d(\xi)\} \equiv \inf\{\xi \geq t : V_B^\xi \leq \mathcal{H}_d\}$), and E_t is the expectation computed under a risk-adjusted measure and under the initial condition $S_t = S$.

The value of a Class A coin can be determined as above recursively, since the holders still get (certain shares of) Class A coin as well as coupons after reset or regular payout. On the right-hand side of (3.1), the first term indicates that on the regular payout date T, holders get a coupon payment RT and a new Class A coin for which the time from the last payment returns to 0 and S is reduced by $RT/2$; the second term indicates that upon upward reset time τ, holders get a coupon payment $R\tau$ and a new Class A coin for which the time from the last payment returns to 0 and S becomes 1; and the third term indicates that upon downward reset time η, holders receive a coupon payment $R\eta$, withdrawal value $1 - |V_B^\eta|$, and $(V_B^\eta)^+$ shares of new Class A coin for which the time from last reset returns to 0 and S becomes 1.

Note that a simulation-based method (e.g., Adams and Clunie (2006)) may not be efficient in achieving real-time calculation of W_A, because the cash flow of Class A coins has an infinite horizon and a weakly path-dependent nature. Therefore, we propose an efficient PDE-based estimation method to compute W_A.

To this end, we assume that P follows a geometric Brownian motion under the risk-adjusted measure:

$$dP_t = rP_t dt + \sigma P_t d\mathcal{B}_t, \tag{3.2}$$

where $\mathcal{B}_t$ is a one-dimensional standard Brownian motion. Under this assumption, since there is no overshoot by a diffusion process, V_B always equals $\mathcal{H}_d$ on a downward reset. Therefore, (3.1) can be simplified to

$$\begin{aligned} W_A(t,S) = E_t\Bigg[& e^{-r(T-t)}(RT + W_A(0, S_T - RT/2)) \cdot \mathbf{1}_{\{T<\tau\wedge\eta\}} + e^{-r(\tau-t)}(R\tau + W_A(0,1)) \cdot \mathbf{1}_{\{\tau\le T\wedge\eta\}} \\ & + e^{-r(\eta-t)}(R\eta + 1 - \mathcal{H}_d + \mathcal{H}_d W_A(0,1)) \cdot \mathbf{1}_{\{\eta\le T\wedge\tau\}} \Bigg]. \end{aligned} \tag{3.3}$$

Theorem 3.1. *Assuming P follows (3.2), W_A is the unique classical solution to the following partial differential equation on $\{(t,S) : 0 \le t < T, H_d(t) < S < H_u(t)\}$ with nonlocal terminal and boundary conditions:*

$$-\frac{\partial W_A}{\partial t} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 W_A}{\partial S^2} + rS\frac{\partial W_A}{\partial S} - rW_A, \qquad 0 \le t < T,\ H_d(t) < S < H_u(t) \tag{3.4}$$

$$W_A(T,S) = RT + W_A(0, S - \frac{1}{2}RT), \qquad H_d(T) < S < H_u(T) \tag{3.5}$$

$$W_A(t, H_u(t)) = Rt + W_A(0,1), \qquad 0 \le t \le T \tag{3.6}$$

$$W_A(t, H_d(t)) = Rt + 1 - \mathcal{H}_d + \mathcal{H}_d W_A(0,1), \qquad 0 \le t \le T. \tag{3.7}$$

By classical solution we mean $W_A \in C^{1,2}(Q) \cap C(\overline{Q}\backslash D)$, where $Q = \{(t,S) : 0 \le t < T, H_d(t) < S < H_u(t)\}$ and $D = \{T\} \times \{H_d(T), H_u(T)\}$.

Theorem 3.1 is a special case of Theorem D.2 in Appendix D of the online supplement. Basically, to show Theorem D.2, one has to first rewrite the expectation in $W_A(t,S)$ in a non-recursive way (see Proposition D.1), and then get the unique PDE representation. The PDE for $W_A(t,S)$ has a unique classical solution thanks to the fact that only the boundary/terminal condition is nonlocal.

The terminal and boundary conditions (3.5) – (3.7) are directly related to the cash flow of Class A coins. Note that these conditions depend on the solution W_A itself. Such nonlocal terminal and boundary conditions make the PDE problem significantly different from the classical Black-Scholes model, leading to challenges in both theory and computation. On the theoretical aspect, due to the nonlocal conditions, the linkage between the stochastic representation (3.3) and the PDE problem (3.4) – (3.7) is not straightforward.[8] On the numerical aspect, the nonlocalness makes the problem nonlinear, which motivates us to propose an efficient iterative procedure to find numerical solutions.

3.2 Numerical Procedure for the Pricing Equation

We propose an iterative algorithm, Algorithm 1, to obtain a numerical solution of the periodic parabolic terminal-boundary value problem (3.4) – (3.7). The convergence of Algorithm 1 is established in Theorem 3.2, whose proof is given in Appendix E.

Algorithm 1: Numerical Procedure for Solving the Pricing Equation

---

initialization: Set initial guess $W_A^{(0)} = 0$ and $i = 0$;
while $||W_A^{(i)} - W_A^{(i-1)}|| >=$ *tolerance* **do**
  $i = i + 1$;
  Given $W_A^{(i-1)}$, solve for $W_A^{(i)}$, the solution to the equation

$$\begin{aligned}
-\frac{\partial W_A}{\partial t} =& \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 W_A}{\partial S^2} + rS\frac{\partial W_A}{\partial S} - rW_A && 0 \le t < T, H_d(t) < S < H_u(t) \\
W_A(1,S) =& RT + W_A^{(i-1)}(0, S - \frac{1}{2}RT) && H_d(t) < S < H_u(t) \\
W_A(t, H_u(t)) =& Rt + W_A^{(i-1)}(0,1) && 0 \le t \le T \\
W_A(t, H_d(t)) =& Rt + 1 - \mathcal{H}_d + \mathcal{H}_d W_A^{(i-1)}(0,1) && 0 \le t \le T.
\end{aligned}$$

end
return: $W_A^{(i)}$

Theorem 3.2. *The sequence $(W_A^{(k)})_{k\ge1}$ defined in Algorithm 1 is monotonically increasing and converges to W_A exponentially; that is, there exists $c_1 > 0$ and $0 < c_2 < 1$ independent of k such that*

$$\sup_{0\le t\le T, H_d(t)\le S\le H_u(t)} |W_A^{(k)}(t,S) - W_A(t,S)| \le c_1 \cdot c_2^k, \text{ as } k \to \infty.$$

Theorem 3.2 implies an exponential convergence rate of Algorithm 1. In Appendix E, we illustrate the convergence using the parameters given in Section 4, where the corresponding c_2 is

[8] In general, there is no explicit solution due to the nonlocal boundary condition. However, due to the linearity of the differential operator, a semi-explicit form may be available in terms of an integral equation.

estimated to be 0.60. The effects of parameters on W_A are illustrated in Appendix F.

3.3 Class A′ and B′ Coins

Denote by $W_{A'}(t, S)$ and $W_{B'}(t, S)$ the market prices of Class A′ and B′ coins, respectively. Since two shares of Class A coin exchange for one A′ coin and one B′ coin, i.e., $2W_A(t, S) = W_{A'}(t, S) + W_{B'}(t, S)$, in the following we only discuss the valuation of Class A′ coin. Under the risk-neutral pricing framework, $W_{A'}(t, S)$ is given recursively as

$$E_t\Bigg[e^{-r(T-t)}(R'T + W_{A'}(0, S_T - RT/2)) \cdot \mathbf{1}_{\{T<\tau\wedge\eta\}} + e^{-r(\tau-t)}(R'\tau + W_{A'}(0,1)) \cdot \mathbf{1}_{\{\tau\le T\wedge\eta\}} + e^{-r(\eta-t)}\left(\min\{R'\eta + 1 - (V_B^\eta)^+, 2(R\eta + 1 + V_B^\eta)\} + (V_B^\eta)^+ W_{A'}(0,1)\right)\mathbf{1}_{\{\eta\le T\wedge\tau\}}\Bigg], \tag{3.8}$$

where τ and η are the first upward and downward reset of Class A (or equivalently, Class A′ and B′) after t, respectively. On a downward reset, if $V_B^\eta > 0$, Class A′ receives coupon $R'\eta$, $1-V_B^\eta$ shares of Class A′ are liquidated, and Class A′ receives the liquidation value; if $\frac{R'\eta-1}{2} - R\eta \le V_B^\eta \le 0$, Class A′ is fully liquidated and still receives full net asset value; otherwise, Class A′ is fully liquidated and takes a loss by receiving $2(1+R\eta+V_B^\eta)$ which is smaller than its net asset value $1+R\eta$. Recall that Class A will suffer a loss if $V_B^\eta < 0$ on a downward reset, therefore Class A′ is safer than Class A since it can still recover its full net asset value in this case, provided $V_B^\eta \ge \frac{R'\eta-1}{2} - R\eta$; only when $V_B^\eta < \frac{R'\eta-1}{2} - R\eta$ will Class A′ take a loss.

Again, assuming that P_t follows a geometric Brownian motion given by (3.2), (3.8) reduces to

$$E_t\Bigg[e^{-r(T-t)}(R'T + W_{A'}(0, S_T - RT/2)) \cdot \mathbf{1}_{\{T<\tau\wedge\eta\}} + e^{-r(\tau-t)}(R'\tau + W_{A'}(0,1)) \cdot \mathbf{1}_{\{\tau\le T\wedge\eta\}} + e^{-r(\eta-t)}\left(R'\eta + 1 - \mathcal{H}_d + \mathcal{H}_d W_{A'}(0,1)\right) \cdot \mathbf{1}_{\{\eta\le T\wedge\tau\}}\Bigg].$$

Similar to the linkage between (3.3) and (3.4)-(3.7), this value can be represented as the unique solution to a PDE (given in Appendix G) and solved numerically via a procedure similar to Algorithm 1.

3.4 The Design with Subsidy from Class A to B

Under the contract design in Section 2.5 where Class A subsidizes Class B on downward resets, one can adapt the risk-neutral valuation formulas for the market value of Class A and A′ . The market value can then be represented as the unique solution to a PDE and solved numerically via a procedure similar to Algorithm 1. The details are given in Appendix H.

4 Numerical Examples

For illustration, we use ETH as the underlying cryptocurrency. The data include the hourly ETH close prices from October 13, 2017 to December 22, 2020. We divide this sample period into two sub-periods: October 13, 2017 to October 12, 2018 for parameter estimation (*Estimation Period* henceforth), and October 13, 2018 to December 22, 2020 for out-of-sample pricing and performance testing (*Testing Period* henceforth). We further assume that the price is monitored on an *hourly* basis, and the upward and downward resets are performed according to the hourly close prices. In practice, the price can be updated at a higher frequency (e.g., minutely) and a reset can occur during the hour once a price update triggers the reset. The default model parameters (annualized) are: $R = 7.30\%$, $R' = 3\%$, $\mathcal{H}_u = 2$, $\mathcal{H}_d = 0.25$, $\tilde{R} = 10\%$, $r = 3\%$, $\sigma = 123\%$, $T = 100$ days. The coupon rates R, R' for Class A and A$'$ mean each share of them will pay coupons with amounts of \$0.02 and \$0.0082 on regular payouts (i.e., for a 100-day period), respectively. The volatility σ is the Black-Scholes volatility estimated from the estimation period. In Appendix I, we demonstrate the robustness of our result against changes in the testing period and estimation period.

4.1 Market Values of Class A and Class B

We first compute the market values of Class A and Class B coins, based on the geometric Brownian motion assumption and the historical prices of ETH. Figure 7 shows that, although Class A has a fixed coupon rate, and its coupon payment is periodic and protected by the resets, its value is still volatile on non-coupon dates. This should be compared to the behavior of a junk bond, whose value is influenced by its issuer's credit risk. In contrast, the main risk of Class A is not credit risk but the risk of a downward reset. On a downward reset, a portion of Class A coins will be liquidated, so the investor will lose the value of future coupons that would be generated from this portion. Therefore, an approaching downward reset will pull down the value of Class A. This is illustrated in the top panel in Figure 7 at the beginning of November 2017: as the downward reset approaches, the value of Class A also goes down, especially when the model underestimates the ETH volatility (by setting $\sigma = 50\%$ p.a.).

The bottom panel of Figure 7 shows the simulated paths from Class B coins. The price volatility of Class B is much higher than that of Class A. Indeed, being a leveraged investment in ETH, the main source of risk for Class B comes from the market risk of ETH. Also, the underlying volatility estimation has a less significant impact on the price volatility of Class B. For both Class A and B, upward reset takes place at 00:00 April 8, 2019, 15:00 May 16, 2019, 17:00 February 12, 2020, 06:00 April 29, 2020, 07:00 July 26, 2020, and 09:00 September 1, 2020. Downward reset takes place at 01:00 November 23, 2018, 18:00 August 28, 2019, and 10:00 March 12, 2020. Regular payout takes place at 01:00 March 3, 2019, 15:00 August 24, 2019, 18:00 December 6, 2019, and 09:00 December 10, 2020.

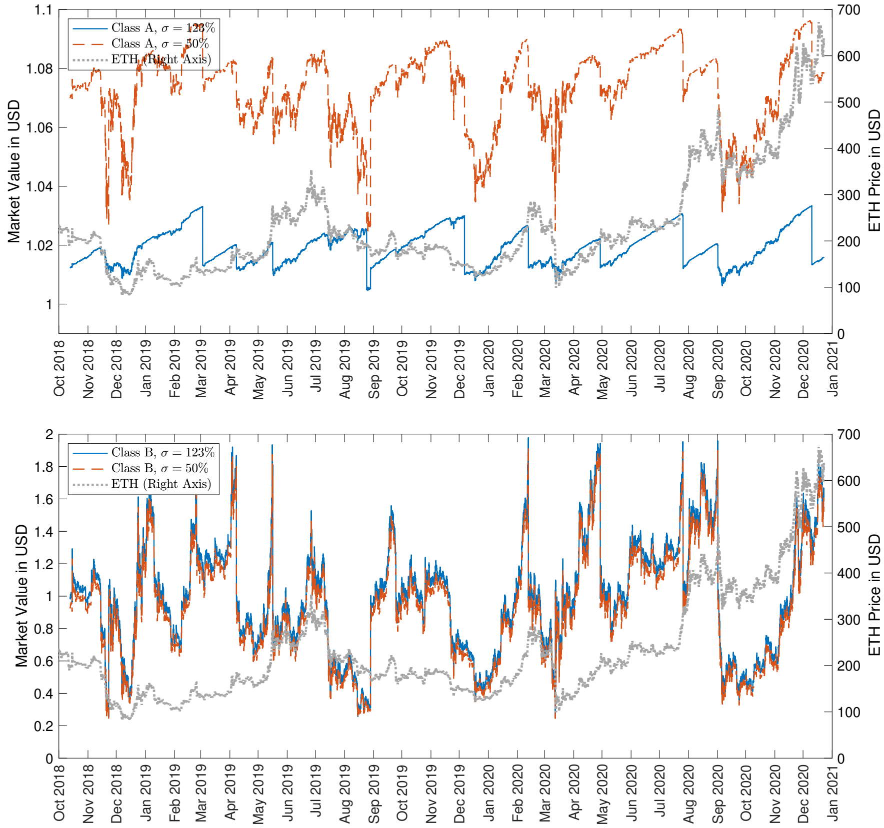

Figure 7: Simulated Class A (top) and Class B (bottom) Market Value per Share in USD, for underlying volatility parameter $\sigma = 123\%$ and 50% p.a. Other parameters (annualized): $R = 7.5\%, \mathcal{H}_d = 0.25$, $\mathcal{H}_u = 2$, $T = 100$, $r = 3\%$. Volatility of Class A market value: 3.48% under $\sigma = 123\%$, and 8.82% under $\sigma = 50\%$.

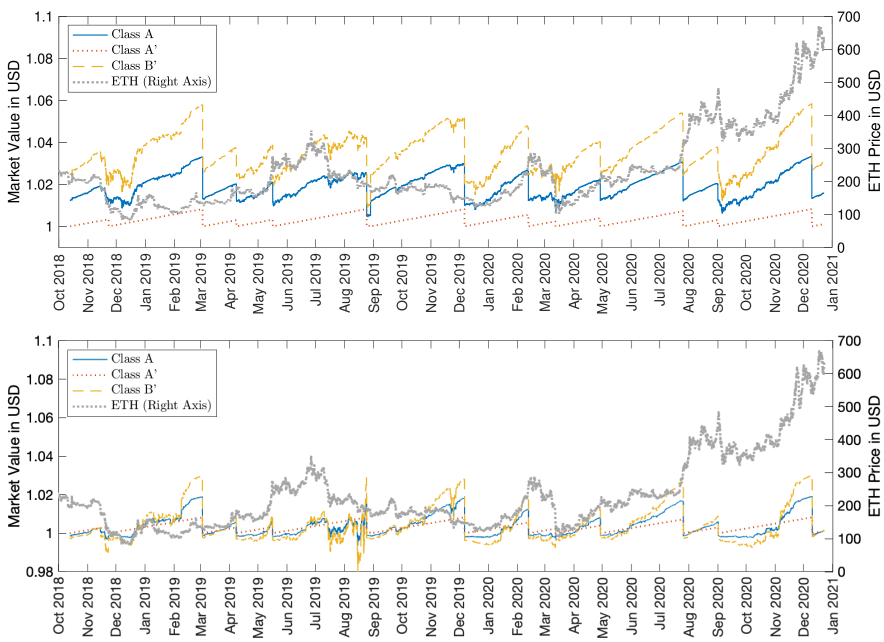

Figure 8: Market Value of Class A$'$ and B$'$ per Share in USD, compared with Class A. **Top figure:** design without subsidy; **Bottom figure:** design with subsidy. Value calculated with estimated volatility $\sigma = 123\%$ p.a. Annualized market value volatility of Class A$'$ and B$'$ is 1.37% and 5.70%, respectively. For the bottom figure, $\tilde{R} = 10\%$. Other parameters (annualized): $R = 7.5\%, \mathcal{H}_d = 0.25$, $\mathcal{H}_u = 2$, $T = 100$, $r = 3\%$.

4.2 Market Value of Class A′ and B′

We can see from Figure 8 that the market value of Class A′ coins is very stable during the testing period (October 13, 2018 to December 22, 2020), with a value close to 1, except for seven downward jumps. These downward jumps correspond to the coupon payment of Class A′ on the reset dates of Class A. Therefore, from the Class A′ holders' point of view, these downward jumps do not necessarily result in a loss due to the delivery of the coupon payments. In comparison, the market value of Class B′ coins is more volatile than Class A, but much less volatile compared to Class B. Indeed, Class B′ represents a leveraged investment in Class A.

By assuming that all coupons are reinvested back into Class A′ coins, the total return of holding Class A′ coins has an annualized volatility of 0.0029%, which is much smaller than that of the total return of holding Class A coins (1.22%) and is even lower than 0.12% of the S&P U.S. Treasury Bill 0–3 Month Index during the same period. This confirms our discussion in Section 2.3 that Class A′ behaves like a money market account; even though the volatility of ETH affects the timing of coupons for Class A′ or partial payback, Class A′ 's stable future cash flow from the linear growth of the accrued coupon results in a minimal impact of ETH volatility on the total return of Class A′ . Even by ignoring the coupon payments and only considering the nominal price, Class A′ still has an annualized return volatility of 1.37%, which is still lower than 5.96% of the USD Index during the same period.

Under the design with a subsidy from Class A to B, the bottom panel of Figure 8 shows that the volatilities of the nominal price of Class A and A′ do not change significantly, being 3.58% and 1.37% (compared to 3.48% and 1.37%), respectively. Furthermore, the volatility of the total return of holding Class A′ also remains the same (0.0029%). The total return volatility of Class A becomes slightly higher at 1.50% (as compared to 1.22% without the subsidy).

5 Jump Risk

In this section, we extend the above geometric Brownian motion-based model (3.2) for the ETH price to a jump-diffusion model to illustrate the performance of the stablecoins in the presence of the event risk (c.f. Liu et al. (2003)).[9] We assume that the ETH price has the following risk-adjusted

[9] One can also incorporate stochastic volatility into the ETH price model, e.g., the Heston model; see also Madan et al. (2019). Compared to a geometric Brownian motion model, this type of model will likely produce a much larger kurtosis for the log return, implying more frequent upward and downward resets. However, stochastic volatility models without jumps exclude black swan events. Incorporating rough volatility models or general jump models for the ETH price is an important direction for future work.

dynamics[10]

$$dP_t/P_{t-} = rdt + \sigma d\mathcal{B}_t + d\sum_{\tau_j \le t}(U_j - 1),$$

where $\{U_j\}$, being nonnegative and i.i.d., are the jump size following distribution ν, and $(\tau_j)_{j\ge1}$ is the sequence of jump times of a Poisson process with intensity λ. We assume that the jump distribution ν is of double-exponential type, that is, $\nu(dz) = \left(p\eta_1 e^{-\eta_1 z}\mathbf{1}_{\{z\ge0\}} + (1-p)\eta_2 e^{\eta_2 z}\mathbf{1}_{\{z<0\}}\right)dz$. This distribution features both a high peak and heavy tails. We estimate the model parameters p, η_1, η_2 by following the two-step procedure in Dang and Forsyth (2016). In the first step, we choose a threshold value k and jointly identify jumps and estimate volatility, where jumps are defined as the returns with an absolute value greater than k times the volatility. In the second step, we fit the jump part and the remaining non-jump part of the returns to the double exponential distribution and normal distribution, respectively. The estimated parameters and the resulting out-of-sample annualized volatility for $k = 2.5, 3.0, 3.5$ are given in Panel A and B of Table 4.

The result shows that, even after factoring in the event-based jump risk, Class A′ coins still have very low volatility. Indeed, even by taking $k = 3.5$, Class A′ under the design without subsidy has a total return volatility of 0.0023%, which is even lower than that under the geometric Brownian motion assumption. This is barely changed by switching to the contract with subsidy.

6 The Impact of Liquidation Cost

So far we have assumed that the coupons, dividends, and partial principal payback are all paid at their equivalent USD values. In practice, liquidating those payments to get USD may be subject to price risk, especially during downward resets when the ETH price is dropping. In this section, we incorporate such liquidation costs into our model and study the stability of A and A′ coins.[11]

For this purpose, we assume that upon a downward reset at time t, the payments from Class A and A′ are subject to a δ fraction haircut; that is, the payments for Class A and A′ are $(1-\delta)(V_A^t - |V_B^t|)$ and $(1-\delta)\min\{1 + R'v_t, 2(1 + Rv_t - |V_B^t|)\}$, respectively. For illustration, we take $\delta = 5\%$. We carry out a stability analysis similar to Panel B of Table 4 and report the result in Panel C under the same parameters. The results show that incorporating the 5% haircut does not have a significant impact on the volatilities of nominal value and total return of both Class A and A′ . For example, the total return volatility of A′ is still very small, around the level of 0.025%.

[10]Even if one does not use the complete hedging argument, using the rational expectations theory, one still gets the same dynamics, except that the risk-free rate r is now endogenously determined in equilibrium by other factors, e.g., an outside endowment process; see, e.g., Kou (2002). More general jump distributions, such as mixed exponential jump-diffusion models in Cai and Kou (2011), can also be potentially used with proper estimation. In fact, one only needs to modify the integral part with respect to the more general jump distribution in the numerical procedure for the partial integro-differential equation. We expect that the main results in this section will still hold qualitatively.

[11]In practice, the cost also includes the processing fees and the cost of congestion in the blockchain, which may not necessarily be proportional to the transaction amount. Our analysis here focuses on the ETH price risk in the process of liquidating the coupons.

Table 4: Parameter Estimation for Jump Diffusion, and Out-of-Sample Annualized Volatility

Panel A: Parameter Estimation

k	p	η_1	η_2	λ	μ	σ
2.5	0.4811	33.6858	32.2112	979	0.6898	0.6679
3.0	0.4237	24.0821	25.4177	465	0.6505	0.8196
3.5	0.4353	18.9006	20.3429	232	0.5944	0.9316

Panel B: No Liquidation Cost

	Contract Design without Subsidy				Contract Design with Subsidy			
	Nominal Value		Total Return		Nominal Value		Total Return	
k	A	A′	A	A′	A	A′	A	A′
2.5	3.2853%	1.3679%	0.3592%	0.0023%	3.3456%	1.3679%	1.1744%	0.0023%
3.0	3.2867%	1.3679%	0.4250%	0.0023%	3.3662%	1.3679%	1.1502%	0.0023%
3.5	3.2918%	1.3679%	0.5107%	0.0023%	3.3932%	1.3679%	1.1457%	0.0023%
No Jump	3.4787%	1.3681%	1.2220%	0.0029%	3.5761%	1.3681%	1.4966%	0.0029%

Panel C: With 5% Liquidation Cost

	Contract Design without Subsidy				Contract Design with Subsidy			
	Nominal Value		Total Return		Nominal Value		Total Return	
k	A	A′	A	A′	A	A′	A	A′
2.5	3.4505%	1.4344%	0.3998%	0.0240%	3.5187%	1.4344%	1.2274%	0.0240%
3.0	3.4538%	1.4347%	0.4686%	0.0247%	3.5408%	1.4347%	1.2042%	0.0247%
3.5	3.4615%	1.4350%	0.5590%	0.0259%	3.5698%	1.4350%	1.2021%	0.0259%
No Jump	3.6717%	1.4345%	1.3180%	0.0430%	3.7704%	1.4345%	1.5914%	0.0430%

Panel A presents the parameter estimation for the double-exponential jump-diffusion model under $k = 2.5, 3, 3.5$, where λ and σ are annualized. Panel B shows the annualized volatility of the nominal value and total return portfolio of Class A and A′ without or with subsidy, under the jump-diffusion model and under the geometric Brownian motion model. Panel C shows the results with 5% liquidation cost as in Section 6. The parameter estimation is based on the hourly returns from October 13, 2017 – October 12, 2018, and the volatility is calculated based on the simulated price of Class A and A′ during the time period October 13, 2018 – December 22, 2020.

7 Comparison Between Class A′ and DAI in Terms of Partial Liquidation

When the ETH price drops, the outstanding number of shares of the stablecoins in both our design and DAI will decrease, due to the downward reset and liquidation event, respectively. In this section, we compare such decreasing effects in our design and DAI. Note that in both DAI and our design, the stability of stablecoins refers to the stability of nominal values, not the market capitalization, due to partial liquidation.

Consider the following scenario as in Klages-Mundt and Minca (2022). There are 100 units of stablecoins created initially (A′ and DAI). For DAI, it is created by locking in $D_0 = \$200 > \$150$ worth of ETH as a deposit. Subsequently, DAI requires that the deposit dollar value must be no less than $\beta = 1.5$ times the unit of DAI created. We assume that the locked-in ETH deposit will not be withdrawn. When the value of the deposit drops below the required value, we assume that the investor will not lock in more ETH as a deposit, and consequently, part of the deposit is liquidated to repay the part of the DAI created. Following the idea in Klages-Mundt and Minca (2022), if at any time $t+1$, the deposit value D_t is below βL_t, where L_t is the remaining quantity of DAI ($L_0 = 100$), then a quantity ℓ_t of DAI is liquidated, such that, post liquidation,

$$D_t - \ell_t = \tilde{\beta}(L_t - \ell_t), \quad \tilde{\beta} = 2,$$

which gives $\ell_t = \frac{\tilde{\beta} L_t - D_t}{\tilde{\beta} - 1}$. We will then focus on L_t, the quantity of DAI remaining at time t. For Class A′ in our design, we focus on the changes in quantity as described in Section 2.3.

We simulate hourly ETH prices based on the dynamics in Section 5 with parameters estimated as in Table 4. For each design, we calculate the average remaining quantity at each time point across all simulated sample paths, which is plotted in Figure 9 (for illustration, we only show the curve until the time point when 20% of the initially created coins remain). On average, more Class A′ remains within a short period after initial issuance, while more DAI remains afterward.

Next, we provide an economic interpretation of the above figure in terms of stochastic dominance. Assuming per unit of stablecoin is liquidated with probability $F(t) = \mathbb{P}[\tau_L \le t]$, the curves in the above figure correspond to $1 - F(t)$, i.e., at time t, $1 - F(t)$ fraction of the initial coins remains. Assuming by holding stablecoins, investors get a constant stream of payoff. Therefore, the total amount of payoff X is proportional to the liquidation time τ_L. Based on this interpretation, we discuss the stochastic dominance between the random payoffs X under Class A′ and DAI. For robustness, we also consider alternative settings with lower market returns, where μ is set to zero or significantly negative while keeping other parameters the same.

Table 5 reports the stochastic dominance (FOSD) relation based on simulation. In terms of first-order stochastic dominance, neither Class A′ nor DAI dominates the other. However, in terms of second-order stochastic dominance (SOSD), the only dominance relation is Class A′ dominates DAI when the hypothetical return $\hat{\mu}$ equals -10 times the estimated return μ. This suggests that in the very extreme case with extremely negative market returns, Class A′ may be preferred to DAI

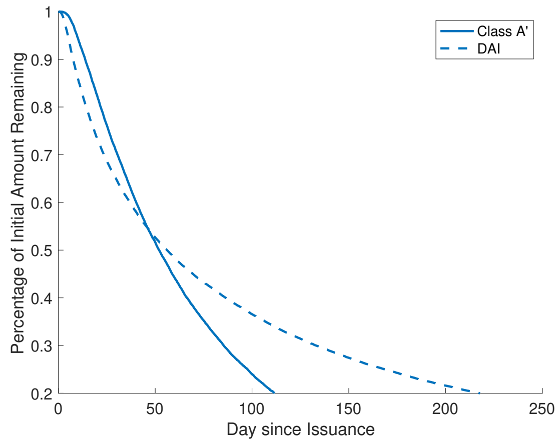

Figure 9: Percentage of Initial Position Remaining After Issuance. Parameters: $k = 2.5, p = 0.4811, \eta_1 = 33.6858, \eta_2 = 32.2112, \lambda = 979, \mu = 0.6898, \sigma = 0.6679$.

by risk-averse investors.

Table 5: Stochastic Dominance between Class A$'$ and DAI

	$k = 2.5$				$k = 3.0$				$k = 3.5$			
$\hat{\mu}$	μ	0	-5μ	-10μ	μ	0	-5μ	-10μ	μ	0	-5μ	-10μ
FOSD	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
SOSD	$\times$	$\times$	$\times$	A$' \succ$ DAI	$\times$	$\times$	$\times$	A$' \succ$ DAI	$\times$	$\times$	$\times$	A$' \succ$ DAI

In this table, $\times$ means neither Class A$'$ nor DAI dominates the other; $\succ$ means stochastic dominant relation. The only stochastic dominant relation in this Table is Class A$'$ is second-order stochastic dominant over DAI when $\mu = -6.8980$. Parameters are as given in Table 4. For each k, in addition to the estimated return μ, we consider three adverse scenarios with hypothetical return $\hat{\mu} = 0, -5\mu, -10\mu$ (other parameters remain the same).

Acknowledgements

Dai acknowledges the supports of NSFC (12071333), Hong Kong GRF (15217123 and 15213422), and The Hong Kong Polytechnic University Research Grants (P0042708, P0042456, P0039114, P0045342, P0046811, and P0051214). Yang acknowledges the supports from the Hong Kong Research Grant Council ECS (24207621) and GRF (14207723).

References

Adams, A. T. and Clunie, J. B. (2006). Risk assessment techniques for split capital investment trusts. *Annals of Actuarial Science*, 1:7–36.

Al-Naji, N. (2018). Basecoin: A price-stable cryptocurrency with an algorithmic central bank. http://www.getbasecoin.com/basecoin_whitepaper_0_99.pdf.

Cai, N. and Kou, S. G. (2011). Option pricing under a mixed-exponential jump diffusion model. *Management Science*, 57(11):2067–2081.

Catalini, C. and Gans, J. S. (2016). Some simple economics of the blockchain. Working Paper.

Chen, S., Chen, C. Y., Härdle, W. K., Lee, T. M., and Ong, B. (2017). A first econometric analysis of the CRIX family. In *Handbook of Blockchain, Digital Finance and Inclusion, Vol 1, Cryptocurrency, FinTech, InsurTech, and Regulation*. Academic Press.

Dai, M., Kou, S., and Yang, C. (2020). A stochastic representation for nonlocal parabolic PDEs with applications. *Mathematics of Operations Research*. forthcoming.

Dai, M., Kou, S., Yang, C., and Ye, Z. (2018). The overpricing of leveraged products: A case study of dual-purpose funds in china. Working Paper.

Dang, D.-M. and Forsyth, P. A. (2016). Better than pre-commitment mean-variance portfolio allocation strategies: A semi-self-financing Hamilton–Jacobi–Bellman equation approach. *European Journal of Operational Research*, 250(3):827–841.

Diem Association (2020). Libra: White paper v2.0. https://www.diem.com/en-us/white-paper/.

Griffin, J. M. and Shams, A. (2020). Is bitcoin really untethered? *The Journal of Finance*, 75(4):1913–1964.

Grinberg, R. (2011). Bitcoin: An innovative alternative digital currency. https://papers.ssrn.com/abstract=1817857.

Harvey, C. R. (2016). Cryptofinance. https://papers.ssrn.com/abstract=2438299.

Hou, A. J., Wang, W., Chen, C. Y., and Härdle, W. K. (2020). Pricing cryptocurrency options. *Journal of Financial Econometrics*, 18(2):250–279.

Ingersoll, J. E. (1976). A theoretical and empirical investigation of the dual purpose funds: An application of contingent-claims analysis. *Journal of Financial Economics*, 3:83–123.

Jarrow, R. A. and O'Hara, M. (1989). Primes and scores: An essay on market imperfections. *The Journal of Finance*, 44:1263–1287.

Kereiakes, E., Do Kwon, M. D. M., and Platias, N. (2019). Terra money: Stability and adoption.

Khapko, M. and Zoican, M. (2018). Smart settlement. https://papers.ssrn.com/abstract=2881331.

Kim, A., Trimborn, S., and Härdle, W. K. (2019). Vcrix-a volatility index for crypto-currencies. *Available at SSRN 3480348*.

Klages-Mundt, A., Harz, D., Gudgeon, L., Liu, J.-Y., and Minca, A. (2020). Stablecoins 2.0: Economic foundations and risk-based models. In *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies*, pages 59–79.

Klages-Mundt, A. and Minca, A. (2022). While stability lasts: A stochastic model of noncustodial stablecoins. *Mathematical Finance*, 32(4):943–981.

Kou, S. G. (2002). A jump diffusion model for option pricing. *Management Science*, 48(8):1086–1101.

Li, Y. and Mayer, S. (2021). Money creation in decentralized finance: A dynamic model of stablecoin and crypto shadow banking. *Fisher College of Business Working Paper*, (2020-03):030.

Liu, J., Longstaff, F. A., and Pan, J. (2003). Dynamic Asset Allocation with Event Risk. *The Journal of Finance*, 58(1):231–259.

Madan, D. B., Reyners, S., and Schoutens, W. (2019). Advanced model calibration on bitcoin options. *Digital Finance*, 1(1):117–137.

MakerTeam (2017). The Dai stablecoin system. https://makerdao.com.

Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system. https://bitcoin.org/bitcoin.pdf.

Rogoff, K. (2015). Costs and benefits to phasing out paper currency. *NBER Macroeconomics Annual*, 29(1):445–456.

Tether (2016). Tether: Fiat currencies on the Bitcoin blockchain. https://tether.to/wp-content/uploads/2016/06/TetherWhitePaper.pdf.

Trimborn, S. and Härdle, W. K. (2018). Crix an index for cryptocurrencies. *Journal of Empirical Finance*, 49:107–122.

Wong, R. et al. (2022). Why stablecoins fail: An economist's post-mortem on terra. *Richmond Fed Economic Brief*, 22(24).

Online Supplement

Designing Stablecoins

A Product Design with General Split Ratio

In Section 2, we have described a specific product design where Class A is stable relative to USD as the target fiat currency and Class B has initial leverage as 2 ($\alpha = 1$). In addition, transaction cost in creation and redemption is omitted. In this section, a general case is discussed.

Dual-class coins can be created by depositing underlying cryptocurrency to the Custodian contract. Upon receiving the underlying cryptocurrency of amount M_C, the Custodian contract will return the sender a certain amount of Class A and Class B coins. Such amount C_A and C_B can be calculated by:

$$C_B = \frac{M_C P_0 \beta_t (1-c)}{1+\alpha}, \quad C_A = \alpha C_B, \tag{A.1}$$

where c is the processing fee of the smart contract, α is a positive number to determine the ratio of A and B, and P_0 is the recorded price of the underlying cryptocurrency in the target fiat currency at the last reset event, and β_t is the conversion factor set as 1 at inception and its behaviour is detailed later in Section A.1 to A.2.

Holders of Class A and Class B coins can withdraw deposited underlying cryptocurrency at any time by performing a redemption. To do this, the user will send αC amount of Class A coins and C amount of Class B coins to the Custodian contract. The contract will deduct Class A and Class B coins and return them to the sender M_C underlying cryptocurrency, where M_C can be calculated by:

$$M_C = \frac{C(1-c)(1+\alpha)}{\beta_t P_0}. \tag{A.2}$$

The net values of coins are calculated based on the coupon rate, the elapsed time from the last reset event, and the latest underlying cryptocurrency price in the target fiat currency fed to the system. In particular:

$$V_A^t = 1 + R \cdot v_t, \quad V_B^t = (1+\alpha) \cdot \frac{P_t}{\beta_t P_0} - \alpha \cdot V_A^t, \tag{A.3}$$

where R is the daily coupon rate, v_t is the number of days from the last reset event, and P_t is the current price of the underlying cryptocurrency in target fiat currency. Note that at all time $Q_A^t = \alpha Q_B^t$, where Q_A^t and Q_B^t are the total amounts of Class A and Class B coins. The implied leverage ratio is

$$L_B^t = \frac{P_t}{\beta_t P_0} \cdot \frac{1+\alpha}{V_B^t}$$

Note that at inception or after contingent resets, the above simply reduces to $L_B^0 = 1 + \alpha$.

A.1 Regular Payout

A regular payout is triggered when $v_{t-} = T$. Upon regular payout:

(i) The total amount of both classes coin remains unchanged, $Q_A^{t+} = Q_A^{t-}$ and $Q_B^{t+} = Q_B^{t-}$

(ii) The net asset value of Class A resets to 1 USD

(iii) Class A holders will receive a certain amount of underlying cryptocurrency from the Custodian contract. Such amount for each Class A coin is $U_A = \frac{V_A^t - 1}{P_t}$

(iv) Conversion factor $\beta_{t+} = \beta_{t-} \cdot \frac{(1+\alpha)P_t}{(1+\alpha)P_t - \alpha\beta_{t-}P_0\left(V_A^t - 1\right)}$

The total value in the system is unchanged after reset:

$$
\begin{aligned}
& U_A \cdot P_t \cdot Q_A^{t-} + Q_A^{t+} \cdot V_A^{t+} + Q_B^{t+} \cdot V_B^{t+} \\
=& \left(V_A^{t-} - 1\right) \cdot Q_A^{t-} + Q_A^{t-} \cdot 1 + V_B^{t-} \cdot Q_B^{t-} \\
=& V_A^{t-} \cdot Q_A^{t-} + V_B^{t-} \cdot Q_B^{t-} \ .
\end{aligned}
$$

A.2 Contingent Upward Reset

An upward reset is triggered when $V_B^t \geqslant \mathcal{H}_u$. Upon upward reset:

(i) The total amount of both classes coins remains unchanged, $Q_A^{t+} = Q_A^{t-}$ and $Q_B^{t+} = Q_B^{t-}$.

(ii) The net asset value of both classes resets to 1 target fiat currency.

(iii) Both classes' holders will receive a certain amount of underlying cryptocurrency from the Custodian contract. Such amount for each Class A coin is $U_A = \frac{V_A^t - 1}{P_t}$ and for each Class B coin is $U_B = \frac{V_B^t - 1}{P_t}$.

(iv) Conversion factor β_{t+} is reset to P_t/P_0.

Total value in the system is unchanged after reset:

$$
\begin{aligned}
& U_A \cdot P_t \cdot Q_A^t + U_B \cdot P_t \cdot Q_B^t + Q_A^{t+} \cdot V_A^{t+} + Q_B^{t+} \cdot V_B^{t+} \\
=& \left(V_A^t - 1\right) \cdot Q_A^t + \left(V_B^t - 1\right) \cdot Q_B^t + Q_A^t \cdot 1 + Q_B^t \cdot 1 \\
=& V_A^t \cdot Q_A^t + V_B^t \cdot Q_B^t \ .
\end{aligned}
$$

A.3 Contingent Downward Reset

A downward reset is triggered when $V_B^t \leqslant \mathcal{H}_d$. Upon downward reset:

(i) The total amount of Class B coins is reduced to $Q_B^{t+} = Q_B^{t-} \cdot (V_B^t)^+$.

(ii) The total amount of Class A coins is reduced to $Q_A^{t+} = Q_B^{t+} \cdot \alpha$.

(iii) The net Value of both classes resets to 1 target fiat currency.

(iv) Class A holders will receive a certain amount of the underlying cryptocurrency from the Custodian contract. Such amount of each Class A coin is: $D_A = \frac{V_A^t - V_B^t}{P_t}$ if $V_B^t \geq 0$, and $D_A = \frac{V_A^t + V_B^t/\alpha}{P_t}$ if $V_B^t < 0$

(v) Conversion factor β_{t+} is reset to P_t/P_0.

The total value in the system is unchanged after reset:

$$
\begin{aligned}
& D_A \cdot P_t \cdot Q_A^{t-} + Q_A^{t+} \cdot V_A^{t+} + Q_B^{t+} \cdot V_B^{t+} \\
= & \left[\left(V_A^t - V_B^t\right) \cdot \mathbf{1}_{V_B^t \geq 0} + \left(V_A^t + V_B^t/\alpha\right) \cdot \mathbf{1}_{V_B^t < 0} \right] \cdot Q_A^{t-} + Q_B^{t+} \cdot \alpha \cdot 1 + Q_B^{t+} \cdot 1 \\
= & V_A^t \cdot Q_A^{t-} - (V_B^t)^+ \cdot Q_B^{t-} \cdot \alpha - (V_B^t)^- \cdot Q_B^{t-} + Q_B^{t-} \cdot (V_B^t)^+ \cdot \alpha + Q_B^{t-} \cdot (V_B^t)^+ \\
= & V_A^t \cdot Q_A^{t-} + V_B^t \cdot Q_B^{t-} \ .
\end{aligned}
$$

Note above used the fact $Q_A^{t-} = Q_B^{t-} \cdot \alpha$.

In the absence of arbitrage, the following price parity shall hold

$$\alpha \cdot W_A^t + W_B^t = \alpha \cdot V_A^t + V_B^t \ ,$$

where W_A^t is the current price of Class A in target fiat currency, and W_B^t is the current price of Class B in target fiat currency.

B Smart Contract Implementation

In order to implement our design, the custodian smart contract performs key tasks, including the creation and redemption of various classes of coins, safekeeping of the underlying digital assets (e.g., ETH), calculating the net values of classes, and execution of reset events. The deposited underlying cryptocurrency is kept by the custodian smart contract. Any user or member of the public can verify the collateral and coins issued through third-party applications such as Etherscan.io. Specifically, Class A and B tokens are implemented as standard ERC20 tokens, which include the standard ERC20 functions:

- `balanceOf()` and `allowance()` for returning the token balance for an address and token allowance for a spender, respectively;
- `transfer()` and `transferFrom()` for token transfer;
- `approve()` for withdrawal approval;
- `totalSupply()` for checking the total token supply.

Specific to the split contract, the implementation includes

- `create()` and `redeem()` for transforming ETH to A and B and vice versa;
- `calculateNav()` for calculating the current net asset values V_A^t and V_B^t;
- `startPreReset()` for checking whether the current values in the system triggers a downward, upward reset or regular payout, based on the rule specified above. If such an event is triggered, it will change the system into a pre-reset state in which A and B are locked to facilitate the reset;
- `startReset()` for executing the upward, downward reset and the regular payout, following the rules as specified in the contract;
- `getSystemAddresses()`, `getSystemStates()`, and `getSystemPrices()` for getting the relative addresses, state parameters, and latest prices;

Besides, `commitPrice()` is used for accepting prices from external price feeds and finding consensus; multiple sources are used for redundancy. The prices are collected hourly as the volume-weighted average of the last five-minute median prices from four exchanges and fed into the smart contract as a fix. At the current capacity, processing 10,000 accounts only takes about 375 seconds, which is very efficient.

The above implementation can be extended in the following ways. First, instead of using only ETH, the underlying can be a diversified portfolio of digital assets, including fiat-backed stablecoins. However, there are two challenges: (a) Currently, major cryptocurrencies (e.g., Bitcoin, Ripple, LiteCoin) can only be transacted on their respective blockchains, and cannot be handled by Ethereum smart contracts; cross-chain projects are still under-developed. (b) A dual-class token on a portfolio of digital assets might be classified as a security in certain countries. Second, one can also use ERC20 tokens on the Ethereum blockchain as the underlying basket. However, these ERC20 tokens are not as popular as the ETH. Third, in the future, it might be possible to run our system on a government-run permitted blockchain if it gains mass adoption. A permitted blockchain could yield higher throughput and shorter block times, which may significantly reduce the time required for resets and support many more accounts.

C Pricing Equations for the Jump Diffusion Models

Following Merton (1976), the pricing PDE of Class A coins, W_A, is given as

$$\begin{aligned}
-\frac{\partial W_A}{\partial t} &= \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 W}{\partial S^2} + (r - \lambda(EU-1))S\frac{\partial W}{\partial S} - rW_A \\
&\quad + \lambda \int_{\mathbb{R}+} W_A(t, Sz)\nu(dz) - \lambda W_A(t,S) \\
W_A(T,S) &= RT + W_A(0, S - \frac{1}{2}RT) && H_d(T) < S < H_u(T) \\
W_A(t,S) &= Rt + W_A(0,1), && 0 \le t \le T,\ S \ge H_u(t) \\
W_A(t,S) &= Rt + 1 - |g(t,S)| + (g(t,S))^+ W_A(0,1), && 0 \le t \le T,\ 0 < S \le H_d(t),
\end{aligned} \tag{C.1}$$

where $g(t,S) = 2S - (1 + Rt)$ is the net asset value of Class B coins. Similarly, the pricing PDE of Class A′ coins, $W_{A'}$, is given as

$$\begin{aligned}
-\frac{\partial W_{A'}}{\partial t} &= \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 W_{A'}}{\partial S^2} + (r - \lambda(EU-1))S\frac{\partial W_{A'}}{\partial S} - rW_{A'} \\
&\quad + \lambda \int_{\mathbb{R}+} W_{A'}(t, Sz)\nu(dz) - \lambda W_{A'}(t,S) \\
W_{A'}(T,S) &= R'T + W_{A'}(0,S) && H_d(T) < S < H_u(T) \\
W_{A'}(t,S) &= R't + W_{A'}(0,S) && 0 \le t \le T, S \ge H_u(t) \\
W_{A'}(t,S) &= \min\{R't + 1 - (g(t,S))^+, 2(Rt + 1 + g(t,S))^+\} \\
&\quad + (g(t,S))^+ W_{A'}(0,1) && 0 \le t \le T, 0 < S \le H_d(t)
\end{aligned} \tag{C.2}$$

The interpretation of the terminal and boundary conditions of W_A and W'_A is similar to that of (3.1) and (3.8), respectively.

The pricing PDEs of Class A and A′ coins with subsidy from A to B are the same as the above two PDEs (replacing W_A and $W_{A'}$ by $\tilde{W}_A$ and $\tilde{W}_{A'}$, respectively), expect that the lower bound conditions (C.1) and (C.2) need to be changed to

$$\tilde{W}_A(t,S) = (Rt - \tilde{R}t + 1 - |g(t,S)|)^+ + (g(t,S))^+ W_A(0,1), \qquad 0 \le t \le T,\ 0 < S \le H_d(t),$$

and

$$\begin{aligned}
W_{A'}(t,S) &= \min\{R't + 1 - (g(t,S))^+, 2(Rt - \tilde{R}t + 1 + g(t,S))^+\} \quad 0 \le t \le T, 0 < S \le H_d(t) \\
&\quad + (g(t,S))^+ W_{A'}(0,1),
\end{aligned}$$

respectively.

D Derivation of the Pricing Equation

Using contract design under general split ratio $\alpha > 0$, similar to (3.1), the value of Class A coins is described by the stochastic representation

$$\begin{aligned} W_A(t,S) = E_t\Bigg[& e^{-r(T-t)}(RT + W_A(0, S_T - \alpha RT/(1+\alpha))) \cdot \mathbf{1}_{\{T<\tau\wedge\eta\}} \\ & + e^{-r(\tau-t)}(R\tau + W_A(0,1)) \cdot \mathbf{1}_{\{\tau\le T\wedge\eta\}} \\ & + e^{-r(\eta-t)}(R\eta + 1 - |V_B^\eta| + (V_B^\eta)^+ W_A(0,1)) \cdot \mathbf{1}_{\{\eta\le T\wedge\tau\}}\Bigg]. \end{aligned} \tag{D.1}$$

Note that (D.1) becomes (3.1) when $\alpha = 1$. In this section, under the geometric Brownian motion assumption, we show that (D.1) defines a unique bounded function W_A, which is exactly the solution to the PDE problem (3.4) – (3.7) when $\alpha = 1$. We denote v_s and Y_s as the time from the last regular payout or reset and the number of A shares at time s, respectively. Starting from an initial value 1, Y is reduced by a factor of $\mathcal{H}_d$ on every downward reset date (thanks to the geometric Brownian motion assumption), reflecting the partial payback of Class A principal. In the remainder of this section, we focus on the right limit process S_{t+}, which is still denoted as S_t for simplicity of notations. Further denote ζ_i, τ_i, and η_i as the i-th regular payout date, upward reset date, and downward reset date after t, respectively. From the construction of contract, we have

$$\begin{aligned} & dS_t = rS_t dt + \sigma S_t d\mathcal{B}_t, \\ & S_{\zeta_i} = S_{\zeta_i -} - \frac{\alpha}{\alpha+1} R v_{\zeta_i -}, \quad S_{\tau_i} = S_{\eta_i} = 1, \quad v_{\tau_i} = v_{\eta_i} = v_{\zeta_i} = 0, \end{aligned}$$

where $\mathcal{B}$ is a Brownian motion under the risk-neutral measure.

First, we derive the following proposition, which expresses the stochastic representation (D.1) into a non-recursive form.

Proposition D.1. *Equation (D.1) defines a unique solution $W_A(t,S)$ for $0 \le t \le T$, $H_d(t) \le S \le H_u(t)$, which can be written as*

$$\begin{aligned} W_A(t,S) = E_t^{(t,S,1)}\Bigg[& \sum_{\zeta_i \ge t} e^{-r(\zeta_i - t)} Y_{\zeta_i -} RT + \sum_{\tau_i \ge t} e^{-r(\tau_i - t)} Y_{\tau_i -} R v_{\tau_i -} \\ & + \sum_{\eta_i \ge t} e^{-r(\eta_i - t)} Y_{\eta_i -}(R v_{\eta_i -} + 1 - \mathcal{H}_d)\Bigg], \end{aligned} \tag{D.2}$$

where $E_t^{(u,s,y)}$ is the $\mathbb{Q}$-expectation computed under the initial condition $v_{t-} = u$, $S_{t-} = s$, and $Y_{t-} = y$.

Note that in (D.2), if t and S are such that t is a regular payout or downward/upward reset date, the right hand side of (3.1) is viewed as the value of the time-t payment plus the expectation with the value of state variables immediately after the jump (if applicable) as time-t starting values.

Proof of Proposition D.1. Without loss of generality we prove the case $T = 1$. We prove this theorem in three steps.

Step 1: To see that W_A given by (D.2) satisfies (D.1), note that (D.2) implies

$$\begin{aligned}
W_A(t,S) = E_t^{t,S,1}\Bigg[& \sum_{t\le\zeta_i<\tau_1\wedge\eta_1} e^{-r(\zeta_i-t)}Y_{\zeta_i-}R + e^{-r(\tau_1-t)}Y_{\tau_1-}Rv_{\tau_1-}\cdot\mathbf{1}_{\{\tau_1<\eta_1\}} \\
& + e^{-r(\eta_1-t)}Y_{\eta_1-}(Rv_{\eta_1-}+1-\mathcal{H}_d)\cdot\mathbf{1}_{\{\eta_1<\tau_1\}} \\
& + e^{-r(\tau_1\wedge\eta_1-t)}Y_{\tau_1\wedge\eta_1}E_{\tau_1\wedge\eta_1}^{(0,1,Y_{\tau_1\wedge\eta_1})}\Bigg(\sum_{\zeta_i\ge\tau_1\wedge\eta_1} e^{-r(\zeta_i-\tau_1\wedge\eta_1)}\frac{Y_{\zeta_i-}}{Y_{\tau_1\wedge\eta_1}}R \\
& + \sum_{\tau_i>\tau_1\wedge\eta_1} e^{-r(\tau_i-\tau_1\wedge\eta_1)}\frac{Y_{\tau_i-}}{Y_{\tau_1\wedge\eta_1}}Rv_{\tau_i-} \\
& + \sum_{\eta_i>\tau_1\wedge\eta_1} e^{-r(\eta_i-\tau_1\wedge\eta_1)}\frac{Y_{\eta_i-}}{Y_{\tau_1\wedge\eta_1}}(Rv_{\eta_i-}+1-\mathcal{H}_d)\Bigg)\Bigg],
\end{aligned}$$

where $E_{\tau_1\wedge\eta_1}^{(u,s,y)}$ denotes the conditional expectation computed at time $\tau_1\wedge\eta_1$ with $(v,S,Y)_{\tau_1\wedge\eta_1} = (u,s,y)$. As a result,

$$\begin{aligned}
& E_{\tau_1\wedge\eta_1}^{(0,1,Y_{\tau_1\wedge\eta_1})}\Bigg[\sum_{\zeta_i\ge\tau_1\wedge\eta_1} e^{-r(\zeta_i-\tau_1\wedge\eta_1)}\frac{Y_{\zeta_i-}}{Y_{\tau_1\wedge\eta_1}}R + \sum_{\tau_i>\tau_1\wedge\eta_1} e^{-r(\tau_i-\tau_1\wedge\eta_1)}\frac{Y_{\tau_i-}}{Y_{\tau_1\wedge\eta_1}}Rv_{\tau_i-} \\
& \quad + \sum_{\eta_i>\tau_1\wedge\eta_1} e^{-r(\eta_i-\tau_1\wedge\eta_1)}\frac{Y_{\eta_i-}}{Y_{\tau_1\wedge\eta_1}}(Rv_{\eta_i-}+1-\mathcal{H}_d)\Bigg] \\
& = E_0^{(0,1,1)}\Bigg[\sum_{\zeta_i\ge 0} e^{-r\zeta_i}Y_{\zeta_i-}R + \sum_{\tau_i\ge 0} e^{-r\tau_i}Y_{\tau_i-}Rv_{\tau_i-} \\
& \quad + \sum_{\eta_i\ge 0} e^{-r\eta_i}Y_{\eta_i-}(Rv_{\eta_i-}+1-\mathcal{H}_d)\Bigg] \\
& = W_A(0,1),
\end{aligned}$$

where the first equality follows from the Markov property of (v,S,Y) and the fact that time 0 cannot be a coupon payment date given $(v,S,Y)_0 = (0,1,1)$. Plugging this equation into the previous equation, we get

$$
\begin{aligned}
W_A(t,S) = E_t^{(t,S,1)} \Bigg[& \sum_{t \le \zeta_i < \tau_1 \wedge \eta_1} e^{-r(\zeta_i - t)} Y_{\zeta_i -} R + e^{-r(\tau_1 - t)} Y_{\tau_1 -} R v_{\tau_1 -} \cdot \mathbf{1}_{\{\tau_1 < \eta_1\}} \\
& + e^{-r(\eta_1 - t)} Y_{\eta_1 -} (R v_{\eta_1 -} + 1 - \mathcal{H}_d) \cdot \mathbf{1}_{\{\eta_1 < \tau_1\}} + e^{-r(\tau_1 \wedge \eta_1 - t)} Y_{\tau_1 \wedge \eta_1} W_A(0,1) \Bigg] \\
= E_t^{(t,S,1)} \Bigg[& \sum_{\zeta_i < \tau_1 \wedge \eta_1} e^{-r(\zeta_i - t)} R + e^{-r(\tau_1 - t)} R(\tau_1 - \lfloor \tau_1 \rfloor + W_A(0,1)) \cdot \mathbf{1}_{\{\tau_1 < \eta_1\}} \\
& + e^{-r(\eta_1 - t)} (R(\eta_1 - \lfloor \eta_1 \rfloor) + 1 - \mathcal{H}_d + \mathcal{H}_d W_A(0,1)) \cdot \mathbf{1}_{\{\eta_1 < \tau_1\}} \Bigg]
\end{aligned}
$$

by the definition of v, Y, and ζ. This yields (D.1).

Step 2: Next we show that any solution W_A satisfying (D.1) is a bounded function of (t, S) in $0 \le t \le 1$, $H_d(t) \le S \le H_u(t)$. Indeed,

$$
\begin{aligned}
W_A(t,S) = E_t^{(t,S,1)} \Bigg[& \sum_{1 \le i < \tau \wedge \eta} e^{-r(i-t)} R + e^{-r(\tau - t)} \Bigg(R(\tau - \lfloor \tau \rfloor) \\
& + W_A(0,1)) \cdot \mathbf{1}_{\{\tau < \eta\}} \\
& + e^{-r(\eta - t)} (R(\eta - \lfloor \eta \rfloor) + 1 - \mathcal{H}_d + \mathcal{H}_d W_A(0,1) \Bigg) \cdot \mathbf{1}_{\{\eta < \tau\}} \Bigg] \\
\le E_t^{(t,S,1)} \Bigg[& \sum_{1 \le i < \tau \wedge \eta} e^{-r(i-t)} R \\
& + e^{-r(\tau \wedge \eta - t)} (R + \max\{W_A(0,1), 1 - \mathcal{H}_d + \mathcal{H}_d W_A(0,1)\}) \Bigg] \\
\le \frac{e^{-r} R}{1 - e^{-r}} & + (R + \max\{W_A(0,1), 1 - \mathcal{H}_d + \mathcal{H}_d W_A(0,1)\}) := \overline{K}.
\end{aligned}
$$

Note that the right-hand side does not depend on t or S.

Step 3: To see the uniqueness, for any W_A satisfying (D.1), by defining the first coupon payment time $\theta_1 = \tau_1 \wedge \eta_1 \wedge 1$, we get

$$\begin{aligned} W_A(t,S) = E_t^{(t,S,1)}\Bigg[e^{-r(\theta_1-t)}\Bigg(& (R + W_A(0, S_{\theta_1-} - \alpha R/(1+\alpha))) \cdot \mathbf{1}_{\{\theta_1<\tau_1\wedge\eta_1\}} \\ & + (R(\theta_1 - \lfloor\theta_1\rfloor) + W_A(0,1)) \cdot \mathbf{1}_{\{\theta_1=\tau_1\}} \\ & + (R(\theta_1 - \lfloor\theta_1\rfloor) + 1 - \mathcal{H}_d + \mathcal{H}_d W_A(0,1)) \cdot \mathbf{1}_{\{\theta_1=\eta_1\}}\Bigg)\Bigg] \\ = E_t^{(t,S,1)}\Bigg[e^{-r(\theta_1-t)}\Bigg(& R\cdot \mathbf{1}_{\{\theta_1=\zeta_1\}} + Rv_{\theta_1-} \cdot \mathbf{1}_{\{\theta_1=\tau_1\}} \\ & + (Rv_{\theta_1-} + 1 - \mathcal{H}_d) \cdot \mathbf{1}_{\{\theta_1=\eta_1\}} + Y_{\theta_1} W_A(v_{\theta_1}, S_{\theta_1})\Bigg)\Bigg]. \end{aligned}$$

Therefore,

$$\begin{aligned} & W_A(t,S) \\ & = E_t^{(t,S,1)}\Bigg[\sum_{\zeta_i\le\theta_1} e^{-r(\zeta_i-t)} Y_{\zeta_i-} R + \sum_{\tau_i\le\theta_1} e^{-r(\tau_i-t)} Y_{\tau_i-} R v_{\tau_i-} \\ & \quad + \sum_{\eta_i\le\theta_1} e^{-r(\eta_i-t)} Y_{\eta_i-}(Rv_{\eta_i-} + 1 - \mathcal{H}_d) + e^{-r(\theta_1-t)} Y_{\theta_1} W_A(0, S_{\theta_1})\Bigg]. \end{aligned}$$

By plugging the expression for $W_A(0,1)$ into the right-hand side and using the Markov property, one gets

$$\begin{aligned} W_A(t,S) = E_t^{(t,S,1)}\Bigg[\Bigg(& \sum_{\zeta_i\le\theta_1} e^{-r(\zeta_i-t)} Y_{\zeta_i-} R + \sum_{\tau_i\le\theta_1} e^{-r(\tau_i-t)} Y_{\tau_i-} R v_{\tau_i-} \\ & + \sum_{\eta_i\le\theta_1} e^{-r(\eta_i-t)} Y_{\eta_i-}(Rv_{\eta_i-} + 1 - \mathcal{H}_d)\Bigg) \\ & + e^{-r(\theta_1-t)} Y_{\theta_1} E_{\theta_1}^{(v_{\theta_1}, S_{\theta_1}, Y_{\theta_1})}\Bigg[\sum_{\theta_1<\zeta_i\le\theta_2} e^{-r(\zeta_i-\theta_1)}\frac{Y_{\zeta_i-}}{Y_{\eta_i}} R + \sum_{\theta_1<\tau_i\le\theta_2} e^{-r(\tau_i-\theta_1)}\frac{Y_{\tau_i-}}{Y_{\theta_1}} R v_{\eta_i-} \\ & + \sum_{\theta_1<\eta_i\le\theta_2} e^{-r(\eta_i-\theta_1)}\frac{Y_{\eta_i-}}{Y_{\theta_1}}(Rv_{\eta_i-} + 1 - \mathcal{H}_d) + e^{-r(\theta_2-\theta_1)}\frac{Y_{\theta_2}}{Y_{\theta_1}} W_A(v_{\theta_2}, S_{\theta_2})\Bigg]. \end{aligned}$$

Thus,

$$\begin{aligned}
&W_A(t,S) \\
&= E_t^{(t,S,1)}\Bigg[\Bigg(\sum_{\zeta_i\le\theta_2} e^{-r(\zeta_i-t)}Y_{\zeta_i-}R + \sum_{\tau_i\le\theta_2} e^{-r(\tau_i-t)}Y_{\tau_i-}Rv_{\tau_i-} \\
&\quad + \sum_{\eta_i\le\theta_2} e^{-r(\eta_i-t)}Y_{\eta_i-}(Rv_{\eta_i-}+1-\mathcal{H}_d) + e^{-r(\theta_2-t)}Y_{\theta_2}W_A(v_{\theta_2},S_{\theta_2})\Bigg)\Bigg].
\end{aligned}$$

Repeating this for N times, we get

$$\begin{aligned}
W_A(t,S) = E_t^{(t,S,1)}\Bigg[&\sum_{t\le\zeta_i\le\theta_N} e^{-r(\zeta_i-t)}Y_{\zeta_i-}R + \sum_{t\le\tau_i\le\theta_N} e^{-r(\tau_i-t)}Y_{\tau_i-}Rv_{\tau_i-} \\
&+ \sum_{t\le\eta_i\le\theta_N} e^{-r(\eta_i-t)}Y_{\eta_i-}(Rv_{\eta_i-}+1-\mathcal{H}_d) + e^{-r(\theta_N-t)}Y_{\theta_N}W_A(v_{\theta_N},S_{\theta_N})\Bigg],
\end{aligned} \tag{D.3}$$

where θ_N denotes the N-th coupon payment time. Thanks to the boundedness of W_A, we have

$$\begin{aligned}
0 &\le \lim_{N\to\infty} E_t^{(t,S,1)}\left[e^{-r(\theta_N-t)}Y_{\theta_N}W_A(v_{\theta_N},S_{\theta_N})\right] \\
&\le \overline{K}\cdot \lim_{N\to\infty} E_t^{(t,S,1)}\left[e^{-r(\theta_N-t)}\right] = 0.
\end{aligned}$$

Here, to see the last equality, it suffices to note that (1) for two consecutive regular payouts ζ_i,ζ_{i+1}, we have $\zeta_{i+1}-\zeta_i \ge T$, therefore $E_t^{t,S,1}\left[e^{-r(\zeta_{i+1}-\zeta_i)}\right] \le e^{-rT} := K_1 < 1$; (2) for two consecutive upward resets τ_i,τ_{i+1}, we have $E_t^{t,S,1}\left[e^{-r(\tau_{i+1}-\tau_i)}\right] = E_0^{0,1,1}\left[e^{-r\tau_1}\right] := K_2 < 1$; (3) for two consecutive downward resets η_i,η_{i+1}, we have $E_t^{t,S,1}\left[e^{-r(\eta_{i+1}-\eta_i)}\right] = E_0^{0,1,1}\left[e^{-r\eta_1}\right] := K_3 < 1$. Consequently, for $k\ge 1$, we have

$$E_t^{t,S,1}\left[e^{-r(\theta_{3k+3}-\theta_{3k})}\right] \le K_1\wedge K_2\wedge K_3 < 1,$$

since there must be two regular payouts, upward resets or downward resets in $\theta_{3k},\ldots,\theta_{3k+3}$. Therefore,

$$E_t^{t,S,1}\left[e^{-r(\theta_{3k+3}-t)}\right] \le e^{rt}\cdot(K_1\wedge K_2\wedge K_3)^k \to 0, \text{ as } k\to\infty, \tag{D.4}$$

and the claim follows from the monotonicity of (θ_k). Therefore, by sending $N\to\infty$, we infer that the right-hand side of (D.3) converges to (D.2). This shows that any W_A is equal to the right-hand side of (D.2), which gives the uniqueness of W_A satisfying (D.1). (D.4) also shows the exponential rate of convergence. □

Theorem D.2. *W_A is the unique classical solution (that is, $W_A \in C^{1,2}(Q)\cap C(\overline{Q}\backslash D)$, where $Q = \{(t,S) : 0\le t<T, H_d(t)<S<H_u(t)\}$ and $D=\{T\}\times\{H_d(T),H_u(T)\}$) to the following*

partial differential equation on $\{(t,S): 0 \le t < T, H_d(t) < S < H_u(t)\}$

$$
\begin{aligned}
-\frac{\partial W_A}{\partial t} &= \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 W_A}{\partial S^2} + rS\frac{\partial W_A}{\partial S} - rW_A, && 0 \le t < T, H_d(t) < S < H_u(t) && \text{(D.5)}\\
W_A(T,S) &= RT + W_A(0, S - \frac{\alpha}{1+\alpha}RT), && H_d(T) < S < H_u(T) && \text{(D.6)}\\
W_A(t, H_u(t)) &= Rt + W_A(0,1), && 0 \le t \le T && \text{(D.7)}\\
W_A(t, H_d(t)) &= Rt + 1 - \mathcal{H}_d + \mathcal{H}_d W_A(0,1), && 0 \le t \le T. && \text{(D.8)}
\end{aligned}
$$

Proof of Theorem D.2. Proposition D.1 shows that we can rewrite (D.1) in a non-recursive form as

$$
\begin{aligned}
W_A(t,S) = E_t^{(t,S,1)}\Bigg[&\sum_{\zeta_i \ge t} e^{-r(\zeta_i - t)} Y_{\zeta_i -} RT + \sum_{\tau_i \ge t} e^{-r(\tau_i - t)} Y_{\tau_i -} R v_{\tau_i -}\\
&+ \sum_{\eta_i \ge t} e^{-r(\eta_i - t)} Y_{\eta_i -}(R v_{\eta_i -} + 1 - \mathcal{H}_d)\Bigg],
\end{aligned}
$$

where $E_t^{(u,s,y)}$ is the $\mathbb{Q}$-expectation computed under the initial condition $v_{t-} = u$, $S_{t-} = s$, and $Y_{t-} = y$. So it remains to show that W_A given as (D.2) is the unique classical solution to (3.4) – (3.7). We prove this result based on the stochastic representation result for nonlocal PDE, i.e., Corollary 3.1 in Dai et al. (2020), and in the following we first establish a connection between this theorem and (D.1).

We first transform S_t to a process $X_t \in [0,1]$:

$$
X_t = \Gamma(v_t, S_t) = \frac{S_t - H_d(v_t)}{H_u(v_t) - H_d(v_t)}.
$$

For X, the lower and upper limit becomes 0 and 1, respectively. X can be interpreted as the relative distance of S to the lower limit H_d in $[H_d(t), H_u(t)]$. Under this transform, by Ito's formula, we have

$$
dX_s = b(v_s, X_s)ds + \sigma(v_s, X_s)d\mathcal{B}_s,
$$

where

$$
\begin{aligned}
b(v,x) &= r(x-1) - \frac{\alpha}{1+\alpha}\frac{R}{H_u(t) - H_d(t)} + \frac{rH_u(t)}{H_u(t) - H_d(t)},\\
\sigma(v,x) &= \sigma\left(\frac{H_u(t)}{H_u(t) - H_d(t)} + x - 1\right).
\end{aligned}
$$

Besides, after this transform, the definition τ_i, η_i and ζ_i becomes

$$\tau_i = \inf\{s > \tau_{i-1} : X_{s-} \geq 1\}, \eta_i = \inf\{s > \eta_{i-1} : X_{s-} \leq 0\}$$
$$\zeta_i = \inf\{s > \zeta_{i-1} : v_{s-} = T, X_{s-} \in (0,1)\}.$$

On these dates, the change of X is described as

$$X_{\zeta_i} = X_{\zeta_i -}, X_{\tau_i} = X_{\eta_i} = \frac{1 - H_d(0)}{H_u(0) - H_d(0)},$$

and on η_i, we have

$$Y_{\eta_i} = \mathcal{H}_d Y_{\eta_i -},$$

due to the reduction in the number of shares.

Now denote $\mathcal{O} = (0,1)$,

$$\begin{aligned}
g(x) &= \frac{1 - H_d(0)}{H_u(0) - H_d(0)} \cdot \mathbf{1}_{x=0,1}(x) + x \cdot \mathbf{1}_{x\in(0,1)}(x) \\
\tilde{\nu}_{t,x} &= \delta_{0,g(x)}(ds, dz) \\
\bar{\nu}(t,x) &= \mathcal{H}_d \cdot \mathbf{1}_{x=0}(x) + 1_{0<x\leq 1}(x) \\
\theta_i &= \inf\{s > \theta_{i-1} : X_{s-} = 0 \text{ or } X_{s-} = 1 \text{ or } v_{s-} = T\} \\
x &= \Gamma(t, S) \\
\tilde{W}_A(t,x) &= W_A(t,S).
\end{aligned}$$

Also, the payouts of Class A coins at regular payout or reset dates can be expressed as $\tilde{h}(v_{\theta_i -}, X_{\theta_i -}, \bar{\nu}(X_{\theta_i -}))$, where $\tilde{h}(v,x,u) = 1 - u + Rv$. Using the above definitions, W_A defined in (D.2) can be expressed as

$$\tilde{W}_A(t,x) = E_t^x \left[\sum_{\theta_i \geq t} e^{-r(\theta_i - t)} Y_{\theta_i -} \tilde{h}(v_{\theta_i -}, X_{\theta_i -}, \bar{\nu}(X_{\theta_i -})) \right].$$

where E_t^x means the expectation calculated under $X_{t-} = x$, $v_{t-} = t$ and $Y_{t-} = 1$. Then, Corollary 3.1 in Dai et al. (2020) shows that $\tilde{W}_A$ is the unique classic solution to

$$-\frac{\partial \tilde{W}_A}{\partial t} - \frac{1}{2}\sigma^2(t,x)\frac{\partial^2 \tilde{W}_A}{\partial x^2} - b(t,x)\frac{\partial \tilde{W}_A}{\partial x} = 0 \qquad \text{in } [0,T) \times (0,1) \tag{D.9}$$
$$\tilde{W}_A(T,x) = RT + \tilde{W}_A(0, g(x)) \qquad \text{in } (0,1) \tag{D.10}$$
$$\tilde{W}_A(t,0) = 1 - \bar{\nu}(0) + Rt + \bar{\nu}(0)\tilde{W}_A(0, g(0)) \qquad \text{on } [0,T] \tag{D.11}$$
$$\tilde{W}_A(t,1) = Rt + \tilde{W}_A(0, g(1)) \qquad \text{on } [0,T]. \tag{D.12}$$

By reverting the transform $(t,s) \mapsto (t,x) = \left(t, \frac{s-H_d(t)}{H_u(t)-H_d(t)}\right)$, we conclude that W_A defined in (D.1) is the unique classical solution to (3.4) – (3.7). $\square$

E Proof of Theorem 3.2

We follow the notations in Section D. From the proof of Theorem D.2, it suffices to consider the sequence $(\tilde{W}_A^{(k)})$ defined via the transformed equation (D.9) – (D.12) by iteration with $\tilde{W}_A^{(0)} = 0$. Indeed, once we show that $(\tilde{W}_A^{(k)})$ converges uniformly to $\tilde{W}_A$, by reverting the transform $(t,s) \mapsto (t,x) = \left(t, \frac{s-H_d(t)}{H_u(t)-H_d(t)}\right)$, we have that $(W_A^{(k)})$ also converges uniformly to W_A, which establishes the theorem. We shall do it in the following two steps.

Step I. First, we show that the solution $\tilde{W}_A^{(k)}(t,x)$ can be stochastically represented as

$$\tilde{W}_A^{(k)}(t,x) = E_t^x \left[\sum_{t \le \theta_i \le \theta_k} e^{-r(\theta_i - t)} Y_{\theta_i -} \tilde{h}(v_{\theta_i -}, X_{\theta_i -}, \bar{\nu}(X_{\theta_i -})) \right].$$

Indeed, for $k=1$, by Feynman-Kac representation for diffusion processes, the equation

$$\begin{aligned}
-\frac{\partial \tilde{W}_A}{\partial t} &= \frac{1}{2}\sigma^2(t,x)\frac{\partial^2 \tilde{W}_A}{\partial x^2} + b(t,x)\frac{\partial \tilde{W}_A}{\partial x} - r\tilde{W}_A && 0 \le t < T, 0 < x < 1\\
\tilde{W}_A(T,x) &= RT && 0 < x < 1\\
\tilde{W}_A(t,1) &= Rt && 0 \le t \le T\\
\tilde{W}_A(t,0) &= Rt + 1 - \mathcal{H}_d && 0 \le t \le T.
\end{aligned}$$

has a unique classical solution $\tilde{W}_A^{(1)} \in C^{1,2}(Q) \cap C(\overline{Q} \backslash D)$, which can be represented as

$$\tilde{W}_A^{(1)}(t,x) = E_t^x \left[e^{-r(\theta_1 - t)} Y_{\theta_1 -} \tilde{h}(v_{\theta_1 -}, X_{\theta_1 -}, \bar{\nu}(X_{\theta_1 -})) \right].$$

This establishes the claim for $k=1$. Now assume that this claim has been established for $k=i$, $i>1$. Then we have

$$\begin{aligned}
&E_t^x \left[\sum_{t \le \theta_j \le \theta_{i+1}} e^{-r(\theta_j - t)} Y_{\theta_j -} \tilde{h}(v_{\theta_j -}, X_{\theta_j -}, \bar{\nu}(X_{\theta_j -})) \right]\\
&= E_t^x \left[e^{-r(\theta_1 - t)} Y_{\theta_1 -} \tilde{h}(v_{\theta_1 -}, X_{\theta_1 -}, \bar{\nu}(X_{\theta_1 -})) \right]\\
&\quad + E_t^x \left[e^{-r(\theta_1 - t)} E_{\theta_1}^{v_{\theta_1}, X_{\theta_1}, Y_{\theta_1}} \left[\sum_{\theta_1 < \theta_j \le \theta_{i+1}} e^{-r(\theta_j - \theta_1)} Y_{\theta_j -} \tilde{h}(v_{\theta_j -}, X_{\theta_j -}, \bar{\nu}(X_{\theta_j -})) \right] \right]\\
&= E_t^x \left[e^{-r(\theta_1 - t)} Y_{\theta_1 -} \tilde{h}(v_{\theta_1 -}, X_{\theta_1 -}, \bar{\nu}(X_{\theta_1 -})) + e^{-r(\theta_1 - t)} Y_{\theta_1} \tilde{W}_A^{(i)}(v_{\theta_1}, X_{\theta_1}) \right],
\end{aligned}$$

where the last equality is from the Markov property and time-homogeneity of (v, X, Y). On the other hand, from the Feynman-Kac representation, the equation

$$\begin{aligned}
-\frac{\partial \tilde{W}_A}{\partial t} =& \frac{1}{2}\sigma^2(t,x)\frac{\partial^2 \tilde{W}_A}{\partial x^2} + b(t,x)\frac{\partial \tilde{W}_A}{\partial x} - r\tilde{W}_A && 0 \le t < T, 0 < x < 1 \\
\tilde{W}_A(T,x) =& RT + \tilde{W}_A^{(i)}(0, g(x)) && 0 < x < 1 \\
\tilde{W}_A(t,1) =& Rt + \tilde{W}_A^{(i)}(0, g(1)) && 0 \le t \le T \\
\tilde{W}_A(t,0) =& Rt + 1 - \mathcal{H}_d + \mathcal{H}_d \tilde{W}_A^{(i)}(0, g(0)) && 0 \le t \le T.
\end{aligned}$$

has a unique classical solution, $\tilde{W}_A^{(i+1)}$, which can be represented as

$$\begin{aligned}
\tilde{W}_A^{(i+1)}(t,x) = E_t^x \Bigg[& e^{-r(\theta_1 - t)} Y_{\theta_1 -} \tilde{h}(v_{\theta_1 -}, X_{\theta_1 -}, \bar{\nu}(X_{\theta_1 -})) \\
& + e^{-r(\theta_1 - t)} Y_{\theta_1} \tilde{W}_A^{(i)}(v_{\theta_1}, X_{\theta_1}) \Bigg].
\end{aligned}$$

Therefore,

$$\tilde{W}_A^{(i+1)}(t,x) = E_t^x \left[\sum_{t \le \theta_j \le \theta_{i+1}} e^{-r(\theta_j - t)} Y_{\theta_j -} \tilde{h}(v_{\theta_j -}, X_{\theta_j -}, \bar{\nu}(X_{\theta_j -})) \right],$$

and the claim is proved by induction.

Step II. Next we prove the uniform convergence of $(\tilde{W}_A^{(k)})$ to $\tilde{W}_A$. To see this,

$$\begin{aligned}
& |\tilde{W}_A^{(k)}(t,x) - \tilde{W}_A(t,x)| \\
=& \left| E_t^x \left[e^{-r(\theta_k - t)} \sum_{\theta_j > \theta_k} e^{-r(\theta_j - \theta_k)} Y_{\theta_j -} \tilde{h}(v_{\theta_j -}, X_{\theta_j -}, \bar{\nu}(X_{\theta_j -})) \right] \right| \\
\le& \bar{K} \cdot \sup_{(t,x) \in [0,T) \times (0,1)} E_t^x \left[e^{-r(\theta_k - t)} \right],
\end{aligned}$$

where $\bar{K}$ is as defined in the proof of Proposition D.1.

We claim that the right-hand side converges to 0. Suppose this claim is proved, we then have the uniform convergence of $(\tilde{W}_A^{(k)})$ in $[0,T) \times (0,1)$. Using the terminal and boundary conditions of the equation for $\tilde{W}_A^{(k)}$, $(\tilde{W}_A^{(k)})$ also converges uniformly on $[0,T] \times [0,1]$.

To verify the claim, it suffices to note that (1) for two consecutive regular payouts ζ_i, ζ_{i+1}, we have $\zeta_{i+1} - \zeta_i \ge T$, therefore $E_t^x \left[e^{-r(\zeta_{i+1} - \zeta_i)} \right] \le e^{-rT} := K_1 < 1$; (2) for two consecutive upward resets τ_i, τ_{i+1}, we have $E_t^x \left[e^{-r(\tau_{i+1} - \tau_i)} \right] = E_0^{g(1)} \left[e^{-r\tau_1} \right] := K_2 < 1$; (3) for two consecutive downward resets η_i, η_{i+1}, we have $E_t^x \left[e^{-r(\eta_{i+1} - \eta_i)} \right] = E_0^{g(0)} \left[e^{-r\eta_1} \right] := K_3 < 1$. Consequently, for $k \ge 1$, we have

$$E_t^x \left[e^{-r(\theta_{3k+3} - \theta_{3k})} \right] \le K_1 \wedge K_2 \wedge K_3 < 1,$$

since there must be two consecutive regular payouts, upward resets or downward resets in $\theta_{3k}, \ldots, \theta_{3k+3}$. Therefore,

$$\sup_{(t,x)\in[0,T)\times(0,1)} E_t^x\left[e^{-r(\theta_{3k+3}-t)}\right] \leq e^{rt} \cdot (K_1 \wedge K_2 \wedge K_3)^k \to 0, \text{ as } k \to \infty,$$

and the claim follows from the monotonicity of (θ_k). $\square$

Using the default parameters in Section 4, we test Algorithm 1 and plot the iteration error $\sup_{0\leq t\leq T, H_d(t)\leq S\leq H_u(t)} |W_A^{(k)}(t,S) - W_A(t,S)|$ with respect to the iteration step k in Figure 10, in which the vertical axis is in logarithmic scale. Figure 10 confirms the exponential rate of convergence, with the corresponding c_2 in Theorem 3.2 estimated to be 0.60.

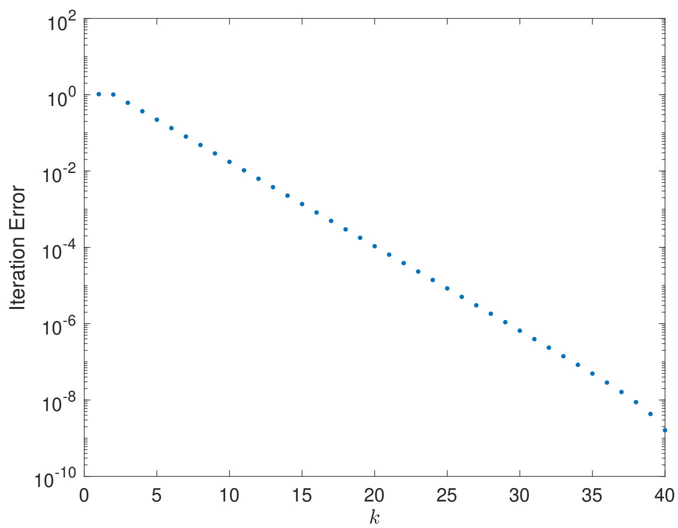

Figure 10: Iteration Errors of Algorithm 1 with Respect to the Step Number k. Parameters are set to their default values given in Section 4.

F Comparative Statics of W_A

In this section, we illustrate the comparative statics of the value of Class A, W_A, under various parameters. The default parameters are as in Section 4. In Figure 11, W_A has an inversed U-shape dependence on S. Indeed, as S decreases towards $\mathcal{H}_d$, the likelihood of a downward reset (and therefore the partial liquidation of Class A) increases, which reduces the future coupon cash flow. This is consistent with the discussion in Section 2.2.4. On the other hand, as S gets closer to $\mathcal{H}_u$, the likelihood of an upward reset increases. This increases the likelihood that S will be reset to the lower value 1 closer to $\mathcal{H}_d$, in spite of the early coupon payment. Subsequently, W_A slightly decreases as S increases towards $\mathcal{H}_u$.

Furthermore, we see that W_A increases with time t from last payment, since a regular payout is closer; W_A increases in the coupon rate R while decreases in the risk-free rate, consistent with the resemblance of Class A and coupon bonds; W_A is smaller when σ or $\mathcal{H}_d$ is larger or when $\mathcal{H}_u$ is smaller, since the likelihood of upward or downward reset is higher in these circumstances.

G Pricing PDE of Class A′

Assuming P follows the geometric Brownian motion (3.2), we can prove similarly as in Theorem 3.1 that $W_{A'}(t, S)$ satisfies the following PDE:

$$
\begin{aligned}
&-\frac{\partial W_{A'}}{\partial t} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 W_{A'}}{\partial S^2} + rS\frac{\partial W_{A'}}{\partial S} - rW_{A'}, && 0 \le t < T, H_d(t) < S < H_u(t) && \text{(G.1)}\\
&W_{A'}(T, S) = R'T + W_{A'}(0, S - \frac{1}{2}RT), && H_d(T) < S < H_u(T) &&\\
&W_{A'}(t, H_u(t)) = R't + W_{A'}(0, 1), && 0 \le t \le T &&\\
&W_{A'}(t, H_d(t)) = R't + 1 - \mathcal{H}_d + \mathcal{H}_d W_{A'}(0, 1), && 0 \le t \le T. && \text{(G.2)}
\end{aligned}
$$

The above periodic PDE can be solved using a similar algorithm as in Theorem 3.2.

H The Design with Subsidy from Class A to B

Under the contract design described in Section 2.5 in which Class A subsidizes Class B on downward resets, one can adapt the risk-neutral valuation formulas for the market value of Class A and A′ . The formulas for the market value of Class A and A′ can be adapted to reflect the difference in the cash flow on downward resets. The market value can then be represented as the unique solution to a PDE and solved numerically via a procedure similar to Algorithm 1.

Specifically, using the same set of notations, the market price of Class A coins with subsidy from A to B, $\tilde{W}_A(t, S)$, can now be written recursively as

$$
\begin{aligned}
E_t\Bigg[& e^{-r(T-t)}[RT + \tilde{W}_A(0, S_T - RT/2)] \cdot \mathbf{1}_{\{T<\tau\wedge\eta\}} + e^{-r(\tau-t)}[R\tau + \tilde{W}_A(0, 1)] \cdot \mathbf{1}_{\{\tau\le T\wedge\eta\}} \\
& + e^{-r(\eta-t)}[(R\eta - \tilde{R}\eta + 1 - |V_B^\eta|)^+ + (V_B^\eta)^+ \tilde{W}_A(0, 1)] \cdot \mathbf{1}_{\{\eta\le T\wedge\tau\}}\Bigg],
\end{aligned} \tag{H.1}
$$

and the market price of Class A′ coins, $\tilde{W}_{A'}(t, S)$, can be written as

$$
\begin{aligned}
E_t\Bigg[& e^{-r(T-t)}(R'T + \tilde{W}_{A'}(0, S_T - RT/2)) \cdot \mathbf{1}_{\{T<\tau\wedge\eta\}} + e^{-r(\tau-t)}(R'\tau + \tilde{W}_{A'}(0, 1)) \cdot \mathbf{1}_{\{\tau\le T\wedge\eta\}} \\
& + e^{-r(\eta-t)} \Big(\min\{R'\eta + 1 - (V_B^\eta)^+, 2(R\eta - \tilde{R}\eta + 1 + V_B^\eta)^+\} + (V_B^\eta)^+ \tilde{W}_{A'}(0, 1)\Big) \mathbf{1}_{\{\eta\le T\wedge\tau\}}\Bigg],
\end{aligned} \tag{H.2}
$$

Assuming that P_t follows a geometric Brownian motion, similar to the linkage between (3.3)

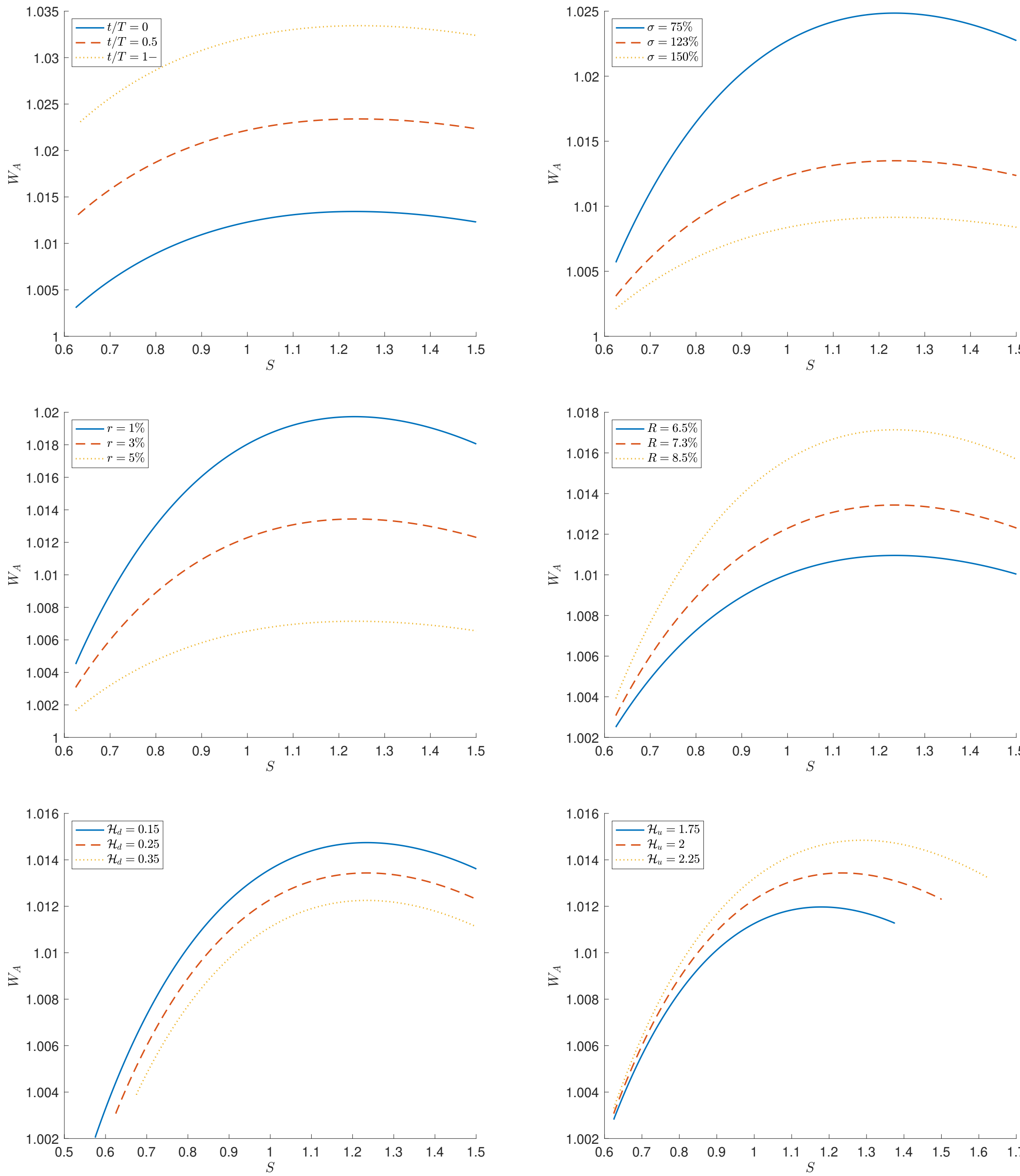

Figure 11: Comparative Statics of the value of Class A W_A. Default parameters (see Section 4): $t = 0$, $T = 100$ days, $R = 7.30\%$,, $\mathcal{H}_u = 2$, $\mathcal{H}_d = 0.25$, $r = 3\%$, $\sigma = 123\%$.

and PDE (3.4) – (3.7), we can prove similarly that $W_A(t,S)$ satisfies the following PDE

$$\begin{aligned}
&-\frac{\partial \tilde{W}_A}{\partial t} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 \tilde{W}_A}{\partial S^2} + rS\frac{\partial \tilde{W}_A}{\partial S} - r\tilde{W}_A, && 0 \le t < T,\ H_d(t) < S < H_u(t) \\
&\tilde{W}_A(T,S) = RT + \tilde{W}_A(0, S - \frac{1}{2}RT), && H_d(T) < S < H_u(T) \\
&\tilde{W}_A(t, H_u(t)) = Rt + \tilde{W}_A(0,1), && 0 \le t \le T \\
&\tilde{W}_A(t, H_d(t)) = (R - \tilde{R})t + 1 - \mathcal{H}_d + \mathcal{H}_d\tilde{W}_A(0,1), && 0 \le t \le T,
\end{aligned}$$

and $W^t_{A'}$ satisfies the PDE (G.1) – (G.2) (after replacing $W_{A'}$ by $\tilde{W}_{A'}$).

I A Check of Robustness: Alternative Starting Date and Parameter Estimation

Our previous numerical test (*Benchmark Test* henceforth) uses a testing period from October 13, 2018 to December 22, 2020 (i.e., assuming that the stablecoin was created on October 13, 2018), with parameters estimated using the estimation period covering the preceding 365 days. One may ask whether the results change by choosing a different creation date. In particular, what happens if the stablecoin is created at a time at which the ETH price is relatively high, such that *the price stays lower than this price for a long period of time*? Also, since Figure 7 suggests that underestimating the underlying volatility can result in higher volatility of Class A, what happens if the parameter is estimated from a period with smaller fluctuation in ETH price, which potentially underestimates the volatility and/or the jump intensity?

To this end, we perform an alternative robustness test (*Robustness Test* henceforth), in which the testing period is from January 13, 2018 to December 22, 2019, and the estimation period is from December 23, 2019 to December 22, 2020. As a result, in the test, the stablecoin is created on January 13, 2018, on which the ETH price reached a peak. Indeed, as shown in Figure 1, the ETH price peaked at \$1421 on January 13, 2018, and this price level has never been recovered subsequently in our dataset. Also, the ETH price is much less volatile during this new parameter estimation period towards the end of our dataset (as confirmed by the lower jump intensity and volatility estimates in Table 6 against those in Table 4). Therefore, choosing this new estimation period can potentially underestimate the ETH volatility. Nevertheless, the resulting volatility of the nominal price of Class A′ , as reported in Table 6, is around 1.31%, which remains close to the value of around 1.37% observed in the benchmark test on the starting date of October 13, 2018. By considering the portfolio value from holding Class A′ coin and reinvesting all dividends, the volatility is around 0.0026%, also close to the level of 0.0023% observed in the benchmark test.

To understand the robustness with respect to a starting date, note that the value of A′ shares depends directly on the relative underlying value $S_t = P_t/(\beta_t P_0)$. By the contract design, the relative value equals 1 at the creation date and jumps back to 1 at every subsequent upward or downward reset. Since the upward and downward resets occur fairly frequently due to the volatile

Table 6: Parameter Estimation for Jump Diffusion, and Out-of-Sample Annualized Volatility, Robustness Test

Panel A: Parameter Estimation

k	p	η_1	η_2	λ	μ	σ
2.5	0.5016	47.3265	44.8230	910	2.0652	0.4771
3.0	0.5105	34.4231	31.0470	381	2.0521	0.5881
3.5	0.4806	27.0859	24.8258	205	2.0379	0.6469

Panel B: Contract Design without Subsidy

	Nominal Value		Total Return	
k	A	A′	A	A′
2.5	2.8174%	1.3112%	0.9401%	0.0025%
3.0	2.8174%	1.3113%	1.0187%	0.0026%
3.5	2.8210%	1.3115%	1.1097%	0.0027%
No Jump	3.8972%	1.3148%	2.7806%	0.0066%

Panel C: Contract Design with Subsidy

	Nominal Value		Total Return	
k	A	A′	A	A′
2.5	2.5719%	1.3112%	1.4015%	0.0025%
3.0	2.5786%	1.3113%	1.4444%	0.0026%
3.5	2.5945%	1.3115%	1.4809%	0.0027%
No Jump	3.7225%	1.3148%	2.7199%	0.0066%

In this table, Panel A presents the parameter estimation for the double-exponential jump-diffusion model under $k = 2.5, 3, 3.5$ under the simulation starting date October 13, 2017; Panel B (resp. Panel C) shows the annualized volatility of the nominal value and total return portfolio of Class A and A′ without (resp. with) subsidy, under the jump-diffusion model and under the geometric Brownian motion model. The parameter estimation is based on the time period December 23, 2019 – December 22, 2020, and the volatility is calculated based on the simulated price of Class A and A′ during the time period October 13, 2017 – December 22, 2019.

nature of ETH prices, the difference between two S paths with different starting dates will eventually be small, even if the ETH price at the starting date is significantly different. In particular, if these two paths share some upward or downward reset date, the relative values from both paths will remain the same after this reset.

Table 7: Liquidation Probability over 10-year Horizon, Robustness Test

	α				
k	0.1	0.2	0.3	0.4	0.5
2.5	0.0000%	0.0000%	0.0000%	0.0001%	0.0145%
3.0	0.0000%	0.0000%	0.0001%	0.0058%	0.1923%
3.5	0.0000%	0.0000%	0.0008%	0.0388%	0.7260%

This table presents the probability of a full liquidation (upon which Class A will have a loss) over a 10-year horizon, for various k and α, under the setting of robustness test.

J The Impact of Split Ratio on Liquidation Probability

In the basic design, one unit of the underlying token can be split into α unit of A and $1-\alpha$ unit of B, where $\alpha \in (0,1)$. A larger α means greater embedded leverage and a higher level of attractiveness of Class B. However, this can also lead to a higher probability of a negative net asset value of Class B upon a downward reset, in which case all classes are fully liquidated, and Class A takes a loss on book value. In this section, we study the impact of α on the probability of such full liquidation events.

We use the jump-diffusion model coefficient estimations reported in Table 4 from the benchmark test and Table 6 from the robustness test. We simulate 10 million ETH price trajectories and calculate the probability that Class A will incur a loss (i.e., a full liquidation) over a 1000-day horizon for various values of α. The results are reported in Panel A of Table 8. With $\alpha = 0.5$, the probability of a full liquidation over the 1000-day horizon ranges from 1.1285% to 6.8080%. By reducing α to 0.3, the full liquidation probability can be reduced below 0.04%.

Table 8: Impact of Split Ratio α

	α				
k	0.1	0.2	0.3	0.4	0.5
Panel A: Liquidation Probability over 1000-day Horizon					
2.5	0.0000%	0.0000%	0.0002%	0.0399%	1.1285%
3.0	0.0000%	0.0000%	0.0057%	0.2683%	3.9588%
3.5	0.0000%	0.0001%	0.0300%	0.7203%	6.8080%
Panel B: Upper Bound UB(α) to Trigger Full Liquidation					
	−67.71%	−48.24%	−35.22%	−25.90%	−18.90%

Panel A of this table presents the probability of a full liquidation (upon which Class A will have a loss) over a 1000-day horizon, for various k and α, based on the jump-diffusion model estimated in Section 5. **Panel B** presents the model-free upper bounds on hourly returns that can trigger a full liquidation for various α. An hourly return above the upper bound will not trigger a full liquidation.

The above liquidation probability is forward-looking in the sense that one extrapolates the historical ETH price data to a 1000-day horizon in the future based on a jump-diffusion model. Now, we will perform an analysis of the full liquidation, which does not depend on any models.

To achieve this, we derive a model-free upper bound for the liquidation probability similar to Section 2.4. According to the contract design, the whole structure of Coin A, B, A′, and B′ will be fully liquidated if and only if the net asset value of B jumps from a level above $\mathcal{H}_d$ to a level below 0. This kind of jump necessarily requires an ETH return lower than the (negative) level of

$$\mathrm{UB}(\alpha) = \max_{0\le v\le T} \frac{-(1-\alpha)\mathcal{H}_d}{\alpha(1+Rv)+(1-\alpha)\mathcal{H}_d}.$$

Panel B of Table 8 reports the model-free upper bounds on the level of hourly ETH price drop necessary to trigger a full liquidation and a loss in Class A, under various values of α. For instance, any hourly (simple) return above −25.90% will not trigger a loss in Class A for $\alpha = 0.4$.

In comparison, the lowest ETH hourly return recorded in our dataset is -19.82%.

This means that by choosing $\alpha = 0.4$ or lower, there will be no full liquidation based on the empirical hourly return distribution in our dataset, independent of any models or any choice of starting date of the testing period. Note that an hourly return below the upper bound does not necessarily lead to a full liquidation. Indeed, there is no full liquidation in our numerical test in Section 4 for $\alpha = 0.5$ despite that the return $-19.82\% \leq -18.90\%$.